



ಬಿ. ಎಂ. ಎಸ್. ತಂತ್ರಿಕ ಮಹಾವಿದ್ಯಾಲಯ
(ಸ್ವಾಯತ್ತ ಹಿರಿಯ ಸಂಸ್ಥೆ)

B.M.S. COLLEGE OF ENGINEERING, BENGALURU - 19
(Autonomous College under VTU)
**Department of CSE (IOT & CYBERSECURITY Including
BLOCKCHAIN TECHNOLOGY)**

**Scheme and Syllabus
III to IV Semester
For the Batch Admitted in the Year 2022**

ಬಿ. ಎಂ. ಎಸ್. ತಂತ್ರಿಕ ಮಹಾವಿದ್ಯಾಲಯ
ಬುಲ್ ಟೆಂಪಲ್ ರೋಡ್ , ಬೆಂಗಳೂರು - 560 019

B.M.S. COLLEGE OF ENGINEERING, BENGALURU - 19
Bull Temple Road, Bengaluru - 560 019



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19
Autonomous Institute, Affiliated to VTU
DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Vision of the Institute

Promoting Prosperity of mankind by augmenting Human Resource Capital through Quality Technical Education Training.

Mission of the Institute

Accomplish Excellence in the field of Technical Education through Education, Research and Service needs of society

Vision of the Department

To be a model centre for education and training in the frontier areas of computing in IoT, Cybersecurity and Blockchain for solving societal problems.

Mission of the Department

Strengthen learning environment that facilitates quality education in the field of IoT, Cybersecurity and Blockchain.

To foster professional ethics and values with concern for environment and society to provide sustainable solutions.

Imbibe passion for research, entrepreneurship, higher studies or professional growth through life-long learning.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19
Autonomous Institute, Affiliated to VTU
DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Typical Credit Distribution *(along with suggested range of credits)*

Curricular Component/ Semester	I	II	III	IV	V	VI	VII	VIII	Total Credits
Basic Science Course (BS)	8	8	3	3	1				23
Engineering Science Course (ES)	9	9	3	3					24
Professional Core Course (PC)			15	14	15	11	2		57
Professional Elective Course (PE)					3	3	3	3	12
Open Elective Course (OE)						3	3	3	9
Project/ Mini-Project (PW)					2	2	6		10
Internship (Re/Ru/In)								10	10
Humanities and Social Sciences, Management Course (HS)	1	1			1				3
Ability Enhancement Course /SDC	2	2	1	1		3	2		11
UHV Courses				1					1
Non-Credit Mandatory Course	-	-	NC	NC	NC	NC	-	-	
Total Credits	20	20	22	22	22	22	16	16	160



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Semester-III

No.	Course Type	Code	Course Title	Credits			Total Credits	Total Hours
				L	T	P		
1	BS	23MA3BSSDM	Statistics & Discrete Mathematics	2	1	0	3	4
2	ES	23DC3ESCOA	Computer Organization & Architecture	3	0	0	3	3
3	PC	23DC3PCDSC	Data Structures	3	0	1	4	5
4	PC	23DC3PCDBM	Database Management Systems	3	0	1	4	5
5	PC	23IC3PCOOP	OOPs with C++	3	0	0	3	3
6	PC	23IC3PCEDS	Embedded Systems	3	0	1	4	5
7	AE	23DC3AEFWD	Full Stack Web Development	0	0	1	1	2
		23IC3AETEW	Technical Writing					
		23IC3AEPMP	Project Management with Git					
8	NCMC		NSS	0	0	0	0	1
			YOGA					
			Physical Edu. (Sports and Athletics)					
	TOTAL			17	1	4	22	28

BS-3, ES-3, PC-15, AE-1



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Semester-IV

Sl.No	Co ur se Ty pe	Code	Course Title	Credits			Total Credit s	Tota l Hour s
				L	T	P		
1	BS	23MA4BSLAO	Linear Algebra and Optimization	2	1	0	3	4
2	ES	23DC4ESTOC	Theory of Computation	3	0	0	3	3
3	PC	23DC4PCOPS	Operating Systems	3	0	0	3	3
4	PC	23DC4PCDAA	Design and Analysis of Algorithms	3	0	1	4	5
5	PC	23IC4PCIOT	Internet Of Things	3	0	0	3	3
6	PC	23IC4PCCNW	Computer Networks	3	0	1	4	5
7	AE	23IC4AEHID	Hardware Interface Design	0	0	1	1	2
		23IC4AEMAD	Mobile Application Development					
		23IC4AEDOP	DevOps					
8	UHV	23MA4AEUHV	Universal Human Values	0	1	0	1	2
9	NCCMC		NSS	0	0	0	0	1
			YOGA					
			Physical Edu. (Sports and Athletics)					
	TOTAL			17	2	3	22	28



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Semester-V

	Course Type	Code	Course Title	Credits			Total Credits	Total Hours
				L	T	P		
1	PC	23IC5PCCSY	Cyber Security	3	0	1	4	5
2	PC	23IC5PCMLG	Machine Learning	3	0	1	4	5
3	PC	23IC5PCCRP	Cryptography	3	1	0	4	5
4	PC	23IC5PCUNP	UNIX System & Network Programming	0	1	0	1	2
5	AE	23DC5AERMI	Research Methodologies and IPR	3	0	0	3	3
6	PE	23IC5PEBDA	Big Data Analytics	3	0	0	3	3
		23IC5PEDSY	Distributed Systems					
		23IC5PEACN	Advanced Computer Network					
7	PW	23IC5PWSEC	Mini Project in Security	0	0	2	2	4
8	HS	23DC5HSEVS	Environmental Studies	1	0	0	1	1
9	NCMC	23NCMC5NS3	NSS	0	0	0	P/NP	1
		23NCMC5YG3	YOGA					
		23NCMC5PE3	Physical Edu. (Sports and Athletics)					
			Details of 40 AICTE Activity Points					
	TOTAL			16	2	4	22	29

PC-12, PE-3, PW-2, HS-2, AE-3



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Semester-VI

	Course Type	Code	Course Title	Credits			Total Credits	Total Hours
				L	T	P		
1	PC	23IC6PCEHG	Ethical Hacking	3	0	1	4	5
2	PC	23IC6PCCCS	Cloud Computing and Security	2	1	0	3	4
3	PC	23IC6PCBCT	Blockchain Technology	3	0	1	4	5
4	PC	23DC6PCSEA	Software Engineering and Agile Methodologies	2	0	0	2	2
5	PE	23IC6PESNA	Social Network Analysis	3	0	0	3	3
		23IC6PEDLN	Deep Learning					
		23IC6PEACR	Advanced Cryptography					
6	OE	23IC6OEEOOP	Object Oriented Programming with C++	3	0	0	3	3
7	PW	23IC6PWPP1	Major Project Phase-1	0	0	2	2	4
8	AE	23IC6AEWEB	Web 3 Development	0	0	1	1	2
		23IC6AEDVZ	Data visualization					
99	NCMC	23NCMC6NS4	NSS	0	0	0	P/NP	1
		23NCMC6YG4	YOGA					
		23NCMC6PE4	Physical Edu. (Sports and Athletics)					
			Details of 60 AICTE Activity Points Earned					
	TOTAL			16	1	5	22	29

PC-13, PE-3,OE-3,PW-2,AE-1



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Sem	III		
Course Title:	Statistics and Discrete Mathematics		
Course Code:	23MA3BSSDM	Total Contact Hours: 39 hours	
L-T-P:	2-1-0	Total Credits:	3

Prerequisites: Basic concepts of Permutations, Combinations, probability, statistics, G.C.D., L.C.D., divisors and primes.

Course Objectives: The goal of the course is to

- Appreciate the importance of Discrete Mathematics and statistics in computer and allied engineering science.
- Acquire the knowledge of Discrete Mathematics and Statistics applied in their core domain.
- Improve their Mathematical Thinking and acquire skills required for sustained lifelong learning.

Teaching-Learning Process (General Instructions)

These are sample Strategies; that teachers can use to accelerate the attainment of the various course outcomes.

- Lecture method (L) does not mean only traditional lecture method, but different types of teaching methods may be adopted to develop the outcomes.
- Show Video/animation films to explain the functioning of various concepts.
- Encourage collaborative (Group Learning) Learning in the class.
- Ask HOT (Higher order Thinking) questions in the class, which promotes critical thinking.
- Adopt Problem Based Learning (PBL), which fosters students' analytical skills, and develops thinking skills such as the ability to evaluate, generalize, and analyze information rather than simply recall it.
- Show the different ways to solve the same problem and encourage the students to come up with their own creative ways to solve them.
- Discuss how every concept can be applied to the real world and when that's possible, it helps to improve the students' understanding.

UNIT-1

GRAPH THEORY

[08 hours]

Basic concepts: Types of graphs, order and size of a graph, in-degree and out-degree, connected and disconnected graphs, Eulerian graphs, Hamiltonian graphs, sub-graphs, and Isomorphic graphs. Matrix representation of graphs: adjacency matrix, incidence matrix. Trees: spanning tree, minimal spanning tree: Kruskal's algorithm and shortest path-Dijkstra's algorithm

Teaching-Learning Process	Chalk and talk method / PowerPoint Presentation
---------------------------	---



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

UNIT-2

PROBABILITY DISTRIBUTIONS

[08 hours]

Theoretical distributions: Discrete and continuous random variables

Discrete distributions: Poisson distribution, Geometric distributions. Continuous distributions: Exponential, Gaussian, Uniform Distribution and Gamma distributions.

Teaching-Learning Process	Chalk and talk method / PowerPoint Presentation
---------------------------	---

UNIT-3

JOINT PROBABILITY AND MARKOV CHAIN

[08 hours]

Joint Probability Distributions: Discrete and continuous joint random variables, Mathematical expectations, Covariance, and Correlation.

Markov Chain: Probability vector, stochastic matrix, fixed point vector, regular stochastic matrix. Higher transition probabilities, stationary distribution of regular Markov chain.

Teaching-Learning Process	Chalk and talk method / PowerPoint Presentation
---------------------------	---

UNIT-4

STATISTICAL INFERENCE

[08 hours]

Introduction, procedure for testing of hypothesis, level of significance.

Large sample: Test of significance for single mean and difference between two means.

Small sample: Test of significance for single mean, the difference between two means, paired t-test, ratio of variances (F-distribution) and Chi-Square goodness of fit.

Teaching-Learning Process	Chalk and talk method / PowerPoint Presentation
---------------------------	---

UNIT-5

COMBINATORICS

[07 hours]

Introduction, Binomial and multinomial theorems, Catalan numbers, the principle of inclusion and exclusion, Derangements, Rook Polynomials.

Teaching-Learning Process	Chalk and talk method / PowerPoint Presentation
---------------------------	---

Course outcomes (Course Skills Set)

After successfully completing the course, the student will be able to understand the topics:

Course Code	CO	COURSE OUTCOME (CO)	PO	Strength
23MA3BSSDM	CO 1	Apply the concept of Discrete Mathematics and Statistics in Computer and Allied Engineering Science.	1	3
	CO 2	Demonstrate the Importance of Discrete Mathematics and Statistics using Modern IT Tools.	1 & 5	3



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Assessment Details (both CIE and SEE)

Component	Type of assessment	Max. Marks	Total	50 % Weightage	Total
CIE – Theory	Quiz	10	100	5	50
	AAT	10		5	
	Test 1	40		20	
	Test 2	40		20	
SEE	End Exam	100		50	

CIE methods /question paper is designed to attain the different levels of Bloom's taxonomy per the outcome defined for the course.

SEMESTER END EXAMINATION:

- Each unit consists of one full question.
- Five full questions to be answered.
- To set one question each from Units 2, 3 & 5 and two questions each from Unit 1 and Unit 4.

Text Books:

1. Graph Theory and Combinatorics, D. S. Chandrasekharaiah, 4th edition, 2011-12, PrismEngineering Education Series.
2. Higher Engineering Mathematics, B. V. Ramana, 2007, Tata McGraw Hill.
3. Discrete Mathematics and its applications, Kenneth H. Rosen, 7th edition, McGraw HillPublishers.

Reference Books:

1. Discrete Mathematics, Kolman, Busby Ross, 5th Edition, 2004, Prentice Hall.
2. Graph Theory with Applications to Engineering and Computer Science, Narsingh Deo, Eastern Economy Edition, PHI Learning Pvt., Ltd.
3. Mathematics for Machine Learning, Marc Peter Deisenroth, A. Aldo Faisal, Cheng Soon Ong, 2020, Cambridge University Press.

E-books and online course materials:

1. <http://jlmartin.faculty.ku.edu/~jlmartin/courses/math725-S16/>
2. https://www.whitman.edu/mathematics/cgt_online/cgt.pdf

Online Courses and Video Lectures:

1. <https://www.coursera.org/learn/probability-intro>
2. [https://nptel.ac.in/courses/111104026/\(DiscreteMathematics\)](https://nptel.ac.in/courses/111104026/(DiscreteMathematics))
3. [https://nptel.ac.in/courses/111106086/\(Combinatorics\)](https://nptel.ac.in/courses/111106086/(Combinatorics))



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Sem	III		
Course Title:	Computer Organization and Architecture		
Course Code:	23DC3ESCOA	Total Contact Hours: 40 hours	
L-T-P:	3-0-0	Total Credits:	3

Unit No.	Topics	Hours
1	Basic Structure of Computers and Instruction Set Architecture: Functional Units, Basic Operational Concepts, Number Representation and Arithmetic Operations, Memory Locations and Addresses, Memory Operations, Instructions, and Instruction Sequencing, Addressing Modes, Stored program concept.	8
2	Introduction to Assembly Language Concepts, Stacks, Subroutines, Additional Instructions, Basic Input/Output: Accessing I/O Devices, Interrupts, Bus Structure, Bus Operation, Arbitration	8
3	Memory System: Basic Concepts, Semiconductor RAM Memories, Read-only Memories, Direct Memory Access, Memory Hierarchy, Cache Memories: Mapping Functions, Virtual Memory	8
4	Arithmetic: Addition and Subtraction of Signed Numbers, Design of Fast Adders, Multiplication of Unsigned Numbers, Multiplication of Signed Numbers Fast Multiplication: Bit-Pair Recoding of Multipliers, Carry-Save Addition of Summands, Summand Addition Tree using 3-2 Reducers, Integer Division, Floating- Point Numbers and Operations: Arithmetic Operations on Floating-Point Numbers, Guard Bits and Truncation, Implementing Floating-Point Operations	8
5	Basic Processing Unit: Some Fundamental Concepts, Instruction Execution, Hardware Components, Instruction Fetch and Execution Steps, Hardwired Control Parallel Computer Architecture: Processor Architecture and Technology Trends, Flynn's Taxonomy of Parallel Architectures, Memory Organization of Parallel Computers: Computers with Distributed Memory Organization, Computers with Shared Memory Organization, Thread-Level Parallelism: Simultaneous Multithreading, Multicore Processors	8

Prescribed Text Book



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Computer Organization and Embedded Systems	Carl Hamacher, Zvonko Vranesic, Safwat Zaky, Naraig Manjikian	6th Edition	McGraw-Hill	12
2.	Parallel Programming for Multicore and Cluster Systems	Thomas Rauber, Gudula Runger	2nd Edition	Springer	13

Reference Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Computer Organization and Design - The Hardware /Software Interface	David A. Patterson, John L. Hennessy	5th Edition	Elsevier	2014
2.	Computer Organization & Architecture	William Stallings	11th Edition	Pearson	2018

MOOC Course				
Sl. No.	Course name	Course Offered By	Year	URL
1.	Computer Architecture and Organization	NPTEL	2022	https://onlinecourses.nptel.ac.in/noc22_cs88/preview

Course Outcomes

At the end of the course the student will be able to

CO1	To apply the concepts of basic functional units to demonstrate the working of computational system.
CO2	To analyze the issues of the processor architecture to improve the efficiency in computer design.
CO3	To design Memory modules and Arithmetic Logic unit for a given specification by analyzing performance issues.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

CO-PO mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2		3										
CO3			2									

Proposed Assessment Plan (for 50 marks of CIE)

Tool	Remarks	Marks
Internals	---	40
QUIZ	---	10
Total		50

SEE Exam Question paper format

Unit-1	Mandatory	One Question to be asked for 20 Marks
Unit-2	Mandatory	One Question to be asked for 20 Marks
Unit-3	Internal Choice	Two Questions to be asked for 20 Marks each
Unit-4	Internal Choice	Two Questions to be asked for 20 Marks each
Unit-5	Mandatory	One Question to be asked for 20 Marks



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19
Autonomous Institute, Affiliated to VTU
DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Semester	III		
Course Title:	Data Structures		
Course Code:	23DC3PCDSC	Total Contact Hours: 40 hours	
L-T-P:	3-0-1	Total Credits:	4

Unit No.	Topics	Hours
1	Introduction To Data Structure: Data Management concepts, Data types – primitive and non-primitive, Types of Data Structures- Linear & Non-Linear Data Structures. Structures and pointers Dynamic memory allocation: allocating a block of memory: Malloc, allocating multiple blocks of memory: Calloc, Releasing the used space: Free Altering the size of memory: Realloc.	8
2	Linear list: Singly linked list implementation, insertion, deletion and searching operations on linear list, circularly linked lists- insertion, deletion and searching operations for circularly linked lists, doubly linked list implementation, insertion, deletion and searching operations, maintaining directory of names, Manipulation of polynomials (addition), representing sparse matrices.	8
3	Stacks: Operations, array representations of stacks, stack applications - infix to postfix conversion, postfix expression evaluation, and function call tracing, recursion. Queues: Introduction, Basic concept, linear queue operations, circular queue, priority queues, double ended queues. Applications of Queues. Stack and queue implementation using linked lists	8
4	Trees: Definitions, tree representation, properties of trees, Binary tree, Binary tree representation, binary tree properties, binary tree traversals, binary tree implementation, Binary Search Tree operations and its implementation, applications of trees.	8
5	Balanced Trees: AVL Trees, Splay trees, Red- Black Trees – Definitions, Rotation and other basic operations.	8



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Prescribed Text Book					
Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Fundamentals of Data Structures in C	Horowitz, Sahni, Anderson Freed	Second	Universities Press	2008
2.	Data Structures using C	Reema Thareja	Second	Oxford University press	2014
Reference Text Book					
Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Data Structures using C	Aaron M. Tenenbaum, Yedidyah Langsam, Moshe J. Augenstein	Fifth	Pearson Education	2007
2	Data Structures - A Pseudocode Approach with C	Richard F. Gilberg Behrouz A. Forouzan	First	Cengage Learning	2005

E-Book						
S l. N o .	Bo ok T itl e	Aut h ors	Ed iti on	Pu blis her	Y e a r	URL
1	Data Structures using C	E. Balagurusamy		Mc Graw Hill	2013	https://dokumen.pub/data-structures-using-c-9781259029547-1259029549.html
2	Data Structures and Program Design in C	Robert L. Kruse, Clovis L. Tondo, Bruce P. Leung	Second	Prentice Hall	1997	https://cdn.preterhuman.net/texts/math/Data_Structure_And_Algorithms/Data%20Structures%20and%20Program%20Design%20in%20C++%20-%20Robert%20L.%20Kruse.pdf



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19
Autonomous Institute, Affiliated to VTU
DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

MOOC Courses				
Sl. No.	Course name	Course Offered By	Year	URL
1	Data Structures	Coursera	2023	https://www.coursera.org/learn/data-structures
2	Data Structures and Algorithms	NPTEL	2023	https://nptel.ac.in/courses/106102064/

Course Outcomes

At the end of the course the student will be able to

CO1	Apply the concept of linear and nonlinear data structures for computing problems.
CO2	Analyse the appropriate data structure operations for a given problem
CO3	Design and develop solutions using the linear and nonlinear data structure for a given specification.
CO4	Conduct experiments for demonstrating the operations of different data structures.

CO-PO mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2		3										
CO3			3									
CO4			3		3					1		



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19
Autonomous Institute, Affiliated to VTU
DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Proposed Assessment Plan (for 50 marks of CIE)

Assessment Tool	No. of Assessments	Marks
Internals	2	20
QUIZ/AAT	1	5
Lab Component	CIE+ Two Lab Tests	25
Total		50

Laboratory Plan

Instructions to Students to be followed in each lab:

1. Each Student should write down the program in the observation book and get it evaluated by the respective lab faculty in-charge and then execute the program.
2. Each Student should bring the lab record with the programs and output written for the programs completed in their respective previous week and get it evaluated by the lab faculty in-charge. In the record book students should - Handwrite the Program - Pasting of the printout of the Output or Handwriting of the Output (Output should be written for all the cases).
3. Students have to practice following list of programs and additional programming exercises will also be given in lab. Students will be made to solve coding challenges on programming platforms like LeetCode and HackerRank.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Lab Program	Unit#	Program Details
1	2	Write a program to implement Singly Linked List with following operations a) Create a linked list. b) Insertion of a node at first position, at any position and at end of list. c) Display the contents of the linked list.
2	2	Write a program to Implement Singly Linked List with following operations a) Create a linked list. b) Deletion of first element, specified element and last element in the list. c) Display the contents of the linked list.
3	2	Write a program to Implement Singly Linked List with following operations a) Sort the linked list. b) Reverse the linked list. c) Concatenation of two linked lists
4	2	Write a program to Implement doubly linked list with primitive operations a) Create a doubly linked list. b) Insert a new node to the left of the node. c) Delete the node based on a specific value d) Display the contents of the list
5	3	Write a program to simulate the working of stack using an array with the following: a) Push b) Pop c) Display The program should print appropriate messages for stack overflow, stack underflow
6	3	Write a program to convert a given valid parenthesized infix arithmetic expression to postfix expression. The expression consists of single character operands and the binary operators + (plus), - (minus), * (multiply) and / (divide)
7	3	Write a program to simulate the working of a queue of integers using an array. Provide the following operations a) Insert b) Delete c) Display



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

		The program should print appropriate messages for queue empty and queue overflow conditions
8	3	Write a program to simulate the working of a circular queue of integers using an array. Provide the following operations. a) Insert b) Delete c) Display The program should print appropriate messages for queue empty and queue overflow conditions
9	3	Write a program to implement Stack & Queues using Linked Representation
10	4	Write a program a) To construct a binary Search tree. b) To traverse the tree using all the methods i.e., in-order, preorder and post order c) To display the elements in the tree.
11	4	Write a program a. To construct a binary search tree b. To implement iterative inorder traversal c. To delete a given element
12	5	Write a program to construct an AVL tree of integers

SEE Question paper format

Unit-1	Mandatory	One Question to be asked for 20 Marks
Unit-2	Internal Choice	Two Questions to be asked for 20 Marks each
Unit-3	Mandatory	One Question to be asked for 20 Marks
Unit-4	Internal Choice	Two Questions to be asked for 20 Marks each
Unit-5	Mandatory	One Question to be asked for 20 Marks



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19
Autonomous Institute, Affiliated to VTU
DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Sem	III		
Course Title:	Database Management Systems		
Course Code:	23DC3PCDBM	Total Contact Hours: 40 hours	
L-T-P:	3-0-1	Total Credits:	4

Unit No.	Topics	Hours
1	Introduction to Database Systems: Introduction, An Example, Characteristics of Database approach, Advantages of using DBMS approach, when not to use a DBMS. Database System Concepts and Architecture: Data models, Schemas and instances, Three schema architecture. SQL: SQL Data Definition and Data Types specifying basic constraints in SQL, Basic retrieval queries in SQL, Insert, Delete and Update statements in SQL, Additional features of SQL, more complex SQL Queries, Specifying Constraints as Assertions and Triggers, Views (Virtual Tables) in SQL, Schema Change Statement in SQL.	8
2	Entity Relation Model: Using High-Level Conceptual Data Models for Database Design, a sample Database Application, Entity types, Entity Sets, Attributes and Keys, Relationship Types, Relationship Sets, Roles and Structural Constraints, Weak Entity types, Refining the ER Design, ER Diagrams, Relationship Types of Degree Higher than two, Relational Database Design using ER to Relational Mapping. Relational Databases: Relational Model Concepts, Relational Model Constraints and Relational Database Schemas, Update Operations, Transactions and Dealing with Constraint Violations, Functional Dependencies	8
3	Relation Algebra: Unary Relational Operations: SELECT and PROJECT, Relational Algebra Operations from Set Theory, Binary Relational Operations: JOIN and DIVISION, Additional Relational Operations, Examples of Queries in Relational Algebra. Normalization: Informal Design Guidelines for Relation Schemas, Functional Dependencies, Normal Forms Based on Primary Keys, General Definitions of Second and Third Normal Forms, Boyce-Codd Normal Form, Multi-valued Dependencies and a Fourth Normal Form, Join Dependencies, Fifth Normal Form.	8

**B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19**

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

4	Transaction Processing Concepts: Introduction to Transaction Processing, Transaction and System Concepts, Desirable Properties of Transactions, Characterizing Schedules Based on Recoverability, Characterizing Schedules Based on Serializability, Transaction Support in SQL, Two-Phase Locking Techniques for Concurrency Control.	8
5	Storage Systems: Overview of Physical Storage Media, Storage Interfaces, Magnetic Disks, Flash Memory, RAID, Disk-Block Access, Database Backup and Recovery from Catastrophic Failures Indexing: Basic Concepts, Ordered Indices, B+-Tree Index Files, B+-Tree Extensions, Hash Indices, Multiple-Key Access, Creation of Indices, Write-Optimized Index Structures, Bitmap Indices, Indexing of Spatial and Temporal Data Query processing & operations	8

Prescribed Text Book					
Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Fundamentals of Database Systems	Elmasri and Navathe	7th Edition	Pearson	2016
2.	Database System Concepts	Silberschatz,H Korth and S Sudarshan	7th Edition	McGrawHill	2019
Reference Text Book					
Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Database Management Systems	Ramakrishnan and Gehrke	3rd Edition	McGrawHill	2014
2.	Database System Design: Design, Implementation, and Management	Peter Rob and Carlos Coronel	8th Edition	CENGAGE Learning	2009



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

E-Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year	URL
1.	An Introduction to Relational Database Theory	Hugh Darwen	3rd Edition	Ventus Publishing ApS	2012	https://www.e-booksdirectory.com/details.php?ebook=3093
2.	Database System The Complete Book	Hector GarciaMolina, JeffreyD. Ullman, Jennifer Widom	Second Edition	Pearson Education	2009	https://people.inf.elte.hu/miiqaai/elektroModulatorDva.pdf

MOOC Course

Sl. No.	Course name	Course offered by	Year	URL
1.	Database Management Systems	SWAYAM	2023	https://onlinecourses.swayam2.ac.in/cec23_cs10/preview
2.	Database Management Essentials	Coursera	2023	https://www.coursera.org/learn/database-management

Course Outcomes

At the end of the course the student will be able to

CO1	Apply the concepts of database management systems for various applications.
CO2	Analyze the given database concepts to its correctness.
CO3	Design and demonstrate conceptual models, query and optimization.
CO4	Ability to conduct experiments to demonstrate the various SQL query processing



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19
Autonomous Institute, Affiliated to VTU
DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

CO-PO mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2		3										
CO3			3									
CO4			3		3							

Proposed Assessment Plan (for 50 marks of CIE)

Tool	Remarks	Marks
Internals	2	25
QUIZ	1	5
Lab Component	CIE + Lab Test	25
Total		50

Laboratory Plan

1. Each Student should write down the work carried out and the outputs in the observation book and get it evaluated by the respective lab faculty in-charge.
2. Students have to practice following SQL queries and additional exercises will also be given in the lab.

	Program Details
1	Sailor Database
2	Supplier Database
3	Salesman Database
4	Movie Database
5	Employee Database



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19
Autonomous Institute, Affiliated to VTU
DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

PROGRAM 1: SAILOR DATABASE

Create tables for the following schema:

SAILOR (sid: integer, sname: string, rating: integer, age: real)

BOAT (bid: integer, bname: string, color: string)

RESERVES (sid: integer, bid: integer, day: date)

Queries:

- 1) Add the required constraints on the created tables.
- 2) Populate the relations with at least 5 tuples each.
- 3) Select names and ages of all sailors. Rename same as 'Sailor Name'
- 4) Find all sailors with a rating above 7
- 5) Find the sid of sailors who have reserved a red boat
- 6) Find the colors of boats reserved by 'Shyam'
- 7) Delete all boats which have never been reserved.

PROGRAM 2: SUPPLIER DATABASE

Create tables for the following schema:

SUPPLIER (sid: integer, name: string, address: string)

PART (pid: integer, name: string, color: string)

CATALOG (sid: integer, pid: integer, cost: real)

Queries:

- 1) Add the required constraints on the created tables.
- 2) Populate the relations with at least 5 tuples each.
- 3) Select the ID and names of all the suppliers.
- 4) Select the most costly part available in the catalog.
- 5) Find the name's of parts for which there is some supplier.
- 6) Find the sids of suppliers who supply a red part and a green part.
- 7) Delete all parts of a given ID.

PROGRAM 3: SALESMAN DATABASE

Create tables for the following schema:

SALESMAN (Salesman_id: integer, Name: string, City: string, Commission: integer)

CUSTOMER (Customer_id: integer, Cust_Name: string, City: string)

ORDERS (Ord_No: integer, Purchase_Amt: real, Ord_Date: date, Customer_id: integer, Salesman_id: integer)

Queries:

- 1) Add the required constraints on the created tables.
- 2) Populate the relations with at least 5 tuples each.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

- 3) Select the ID and names of all the customers.
- 4) Select the salesman with the highest commission.
- 5) List all the orders placed in descending order of their purchase amount.
- 6) Select customers who have salesmen in their cities.
- 7) Delete all orders placed before Jan 2018.

PROGRAM 4: MOVIE DATABASE

Consider the schema for Movie Database:

ACTOR(Act_id, Act_Name, Act_Gender)

DIRECTOR(Dir_id, Dir_Name, Dir_Phone)

MOVIES(Mov_id, Mov_Title, Mov_Year, Mov_Lang, Dir_id)

MOVIE_CAST(Act_id, Mov_id, Role)

RATING(Mov_id, Rev_Stars)

Queries:

- i. List the titles of all movies directed by 'Hitchcock'.
- ii. Find the movie names where one or more actors acted in two or more movies.
- iii. List all actors who acted in a movie before 2000 and also in a movie after 2015 (use JOIN operation).
- iv. Find the title of movies and number of stars for each movie that has at least one rating and find the highest number of stars that movie received. Sort the result by movie title.
- vi. Update rating of all movies directed by 'Steven Spielberg' to 5.

PROGRAM 5: EMPLOYEE DATABASE

Create the following Tables:

LOCATION	
Location_ID	Regional_Group
122	NEW YORK
123	DALLAS
124	CHICAGO
167	BOSTON

DEPARTMENT		
Department_ID	Name	Location_ID
10	ACCOUNTING	122
20	RESEARCH	124
30	SALES	123
40	OPERATIONS	167



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

JOB	
Job_ID	Function
667	CLERK
668	STAFF
669	ANALYST
670	SALESPERSON
671	MANAGER
672	PRESIDENT

EMPLOYEE									
EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MIDDLE_NAME	JOB_ID	MANAGER_ID	HIRE_DATE	SALARY	COMMITMENT	DEPARTMENT_ID
7839	MEGAN	JOHN	S	672	NULL	12-DEC-14	5500	NULL	30
7369	SMITH	JOHN	Q	667	7521	17-DEC-18	800	NULL	20
7499	ALLEN	KEVIN	J	670	7507	20-FEB-17	1600	300	30
7505	DOYLE	JEAN	K	671	7839	04-APR-15	2850	NULL	30
7506	DENNIS	LYNN	S	671	7839	15-MAY-15	2750	NULL	30
7507	BAKER	LESLIE	D	671	7839	10-JUN-15	2200	NULL	40
7521	WARK	CYNTIA	D	670	7505	22-FEB-15	1250	500	30



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19
Autonomous Institute, Affiliated to VTU
DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Queries based on the above tables:

Order By Clause:

1. List out the employee id, last name in ascending order based on the employee id.
2. List out the employee id, name in descending order based on salary column

Group By & Having Clause:

3. How many employees who are working in different departments wise in the organization
4. List out the department wise maximum salary, minimum salary, average salary of the employees
5. List out the job wise maximum salary, minimum salary, average salaries of the employees.
6. List out the no.of employees joined in every month in ascending order.
7. How many employees joined in 1985?
8. How many employees joined in March 1985.
9. Which is the department id, having greater than or equal to 3 employees joined in April 1985.

Sub-Queries

10. Display the employee who got the maximum salary.
11. Display the employees who are working in Sales department
12. Display the employees who are working as "Clerk".
13. Display the employees who are working in "New York"
14. Find out the number of employees working in the "Sales" department.
15. Delete the employees who are working in the accounting department.
16. Display the second highest salary drawing employee details.

Subquery operators: (ALL, ANY, SOME, EXISTS)

17. List out the employees who earn more than every employee in department 30.
18. List out the employees who earn more than the lowest salary in department 30.
19. Find out which department does not have any employees.

Simple join

20. List our employees with their department names
21. Display employees with their designations (jobs)
22. How many employees are working in the sales department?

Non – Equi Join:

23. Display employee details with salary grades.
24. List out the no. of employees on grade wise.

Self-Join:

25. Display the employee details with their manager names.
26. Display the employee details who earn more than their manager's salaries.

Outer Join:

27. Display employee details with all departments.
28. Display all employees in sales or operation departments.

Set Operators:



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

29. List out the distinct jobs in Sales and Accounting Departments.
30. List out the ALL jobs in Sales and Accounting Departments.
31. List out the common jobs in Research and Accounting Departments in ascending order.

SEE Exam Question paper format

Unit-1	Mandatory	One Question to be asked for 20 Marks
Unit-2	Internal Choice	Two Questions to be asked for 20 Marks each
Unit-3	Internal Choice	Two Questions to be asked for 20 Marks each
Unit-4	Mandatory	One Question to be asked for 20 Marks
Unit-5	Mandatory	One Question to be asked for 20 Marks



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Sem	III		
Course Title:	Object Oriented Programming with C++		
Course Code:	23IC3PCOOP	Total Contact Hours: 40 hours	
L-T-P:	3-0-0	Total Credits:	3

Unit No.	Topics	Hours
1	Introduction: Overview to Object Oriented Programming, Benefits and applications of Object-Oriented Programming. Beginning with C++: Program structure, cin and cout objects, namespace, identifiers, variables, constants, operators, reference types, typecasting, control structures. Objects and Classes: Basics of object and class in C++, private and public members, specifying a class, C++ program with a class, arrays within a class, memory allocation to objects, Array of objects.	8
2	Functions in C++: Functions, Inline Function, function overloading, Default arguments, Friend functions, static data and function members, Objects as function arguments, returning objects, constant member functions and objects, this pointer. Constructors and Destructors: Constructors and its types – Default constructors, Parameterized constructors, multiple constructors in a class, Constructors with default arguments, dynamic initialization of objects, Copy constructor, Dynamic constructors, Destructors.	8
3	Operator overloading and Type conversion: Defining Operator overloading, Overloading Unary operators, Binary Operators using friend functions and Member functions, Overloading Special operators , Comma operator, new and delete operators, Rules for overloading operators, Type conversion. Inheritance: Concept of Inheritance, types of inheritance - single, multiple, multilevel, hierarchical, hybrid, protected members, overriding, virtual base class.	8
4	Polymorphism: Pointers, Pointers to Objects, pointers to members and member functions, Pointers to derived classes, virtual and pure virtual functions. I/O and File Management: Concept of streams, C++ stream classes, Unformatted and formatted I/O, manipulators, C++ File stream classes, File management functions, File modes, Binary and Random Files.	8



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

5	Templates: Introduction to templates, function templates and class templates. Introduction to exception: try-catch throw, multiple catch, catch all, rethrowing exception, User defined exceptions. Introduction to Standard Template Library: Components of STL - Containers, Algorithms, Iterators, Application of Container classes	8

Prescribed Text Book					
Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Object Oriented Programming with C++	E Balaguruswamy	6th Edition	Tata McGraw Hill	13
2.	C++ - The Complete Reference	Herbert Schildt	4th Edition	Tata McGraw Hill	2015
Reference Text Book					
Sl. No.	Book Title	Authors	dition	Publisher	Year
1.	C++ Primer	Stanley B Lippman, Josee Lajoie, Barbara E Moo	6th Edition	Addison Wesley Professional	-
2.	Object-Oriented Programming in C++	Robert Lafore	4th Edition	Pearson Education.	-

MOOC Course				
Sl. No.	Course name	Cours e Offere d By	Yea r	URL
1.	Programming in C++	NPTEL	2021	https://onlinecourses.nptel.ac.in/noc21_cs02/preview
2.	IITBombayX : Object- Oriented Programming	edX	2022	https://www.edx.org/course/object-oriented-programming-2



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19
Autonomous Institute, Affiliated to VTU
DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Course Outcomes

At the end of the course the student will be able to

CO1	Apply the object-oriented concepts to find the solution for a given problem.
CO2	Analyse the given application for correctness/functionalities.
CO3	Develop C++ programs / applications for a given requirement.

CO-PO mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	O10	O11	O12
CO1	3											
CO2		3										
CO3			3		1					1		

Proposed Assessment Plan (for 50 marks of CIE)

Tool	Remarks	Marks
Internals	---	40
QUIZ	---	10
Total		50

SEE Exam Question paper format

Unit-1	Mandatory	One Question to be asked for 20 Marks
Unit-2	Mandatory	One Question to be asked for 20 Marks
Unit-3	Internal Choice	Two Questions to be asked for 20 Marks each
Unit-4	Internal Choice	Two Questions to be asked for 20 Marks each
Unit-5	Mandatory	One Question to be asked for 20 Marks



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Sem	III		
Course Title:	Embedded System		
Course Code:	23IC3PCEDS	Total Contact Hours: 40 hours	
L-T-P:	3-0-1	Total Credits:	4

Unit No.	Topics	Hrs
1	Introduction to Internet of Things - Introduction, Definition & Characteristics of IoT, Physical Design of IoT- Things in IoT, IoT Protocols Logical Design of IoT-IoT Functional Blocks, IoT Communication Models, IoT Communication APIs IoT Enabling Technologies- Wireless Sensor Networks, Cloud Computing, Big Data Analytics, Communication Protocols, Embedded Systems IoT Levels & Deployment Templates- IoT Level-1, IoT Level-2 IoT Level-3, IoT Level-4, IoT Level-5, IoT Level-6	8 Hrs
2	Arduino Programming Introduction to Arduino – Types of Arduino – Arduino Toolchain – Arduino Programming Structure – Sketches – Input/Output from Pins Using Sketches. Introduction to Sensors and Actuators- Classification and Selection Criteria.	8 Hrs
3	Prototype development with Arduino Integration of Sensors and Actuators with Arduino. Setting Up Arduino- for integrating various sensors and controlling Relay, controlling Servo Motor, playing Sound, using an Alphanumeric LCD .	8 Hrs



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

4	Prototype development with Raspberry Pi Raspberry Pi- Board details, Linux on Raspberry Pi, Raspberry Pi Interfaces, Programming Raspberry Pi with Python, Other IoT devices.	8 Hrs
5	IoT Applications- Case studies - Home Automation, Cities, Environment, Energy, Retail, Logistics, Agriculture, Industry, Health and Lifestyle	8 Hrs

Prescribed Text Book

Sl.No	Book Title	Authors	Edition	Publisher	Year
1.	Internet of Things: A HandsOn Approach	Arshdeep Bahga, Vijay Madiseti	1 st	Orient Blackswan Private Limited	2015
2.	Internet of Things	Srinivasa K G, Siddesh G.M., Hanumantha Raju R.	1st	Cengage Learning India Pvt. Ltd.	2017
3.	Internet of Things: Architecture and Design Principles	Raj Kamal	1 st	McGraw Hill Education	2017

E-Book

Sl. No	Book Title	Authors	Edition	Publisher	Year	URL
1.	Designing for the Internet of Things	---	2	O'reilly	2017	https://www.oreilly.com/design/free/designing-for-the-internet-of-things.csp
2.	Using the Web to Build the IoT	DOMINIQUE GUINARD	2	Manning Publisher	2016	https://webofthings.org/2016/04/24/free-e-book-using-the-web-to-build-the-iot/



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

MOOC Course				
Sl. No.	Course name	Course Offered By	Year	URL
1.	Introduction to Internet of Things	IIT Kharagpur	2018	https://nptel.ac.in/courses/106/105/106105166/
2.	AWS IoT: Developing and Deploying an Internet of Things	Edx	2020	https://www.edx.org/course/aws-iot-developing-and-deploying-an-internet-of-th

Course Outcomes

At the end of the course the student will be able to

CO1	Apply various sensors and actuators for designing IoT applications using arduino programming.
CO2	Analyze the various IoT architectural components.
CO3	Design IOT systems using the Arduino development board and Raspberry Pi by interfacing sensors, communication modules and actuators.

CO-PO mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	O10	O11	2
CO1	2		2									
CO2		2										
CO3	1	1	3		1							



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19
Autonomous Institute, Affiliated to VTU
DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Proposed Assessment Plan (for 50 marks of CIE)

Tool	Remarks	Marks
Internals	---	20
AAT	---	05
Lab		25
Total		50

SEE Exam Question paper format

Unit-1	Mandatory	One Question to be asked for 20 Marks
Unit-2	Mandatory	One Question to be asked for 20 Marks
Unit-3	Internal Choice	Two Questions to be asked for 20 Marks each
Unit-4	Internal Choice	Two Questions to be asked for 20 Marks each
Unit-5	Mandatory	One Question to be asked for 20 Marks



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19
Autonomous Institute, Affiliated to VTU
DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Internet of Things Lab - Plan of Activities

Evaluation: 25 Marks

Instructions to Students to be followed in each IOT lab:

1. Each Student should write down the work carried out and the outputs in the observation book and get it evaluated by the respective lab faculty in-charge.
2. Each Student should bring the lab record with the programs and output written for the programs completed in their respective previous week and get it evaluated by the lab faculty in-charge. In the record book students should
 - Handwrite the Circuit diagram
 - Handwrite the Program
3. Each Student should practice programs using different sensor and actuator combinations also.

Note: Lab test will consist of a new set of programs, but designed using the sensor and actuators practiced in the lab.

Experiment #	Name of Experiment
1	Digital and Analog output interfacing with Arduino Board
2	Interfacing Push button ,potentiometer with Arduino Board
3	Design and implement the turning ON/OFF of light based on surrounding light intensity using LDR.
4	Design and implement Fire alarm system using flame sensor and buzzer
5	Design and implement smart irrigation system using Soil Moisture sensor and Servo Motor
6	To design and implement an obstacle detection system using ultrasonic sensors and LCD.
7	Design and implement Color recognizing system using color sensor and RGB LED.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

8	Interface motion sensor with Arduino Uno board
9	Study of Connectivity and Configuration of Raspberry Pi with basic peripherals, Understanding GPIO and its use in programs.
10	Write a program on Raspberry Pi to detect an obstacle using ultrasonic sensor.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Sem	III		
Course Title:	Full Stack Web development		
Course Code:	23DC3AEFWD	Total Contact Hours: 40 hours	
L-T-P:	0-0-1	Total Credits:	1

Introduction

Database Application Development - Under this project work, student should develop back end data base table for any chosen data base applications.

Under this project work, student should develop Advanced Web based Application using technologies such as PHP, Python, Node JS, React, Angular.

Students can form a group with minimum of two and maximum of four.

Teacher allotted for project work to students should teach full stack technologies like Node JS, React, etc., during Class/Lab hours as per the allotment. Teacher allotted for project work should guide the students in choosing the topic and towards carrying out project work and complete the evaluation of assigned students.

Teacher allotted for project work to students should teach devops tools like git, github, Jenkins, docker, etc.

Course Outcomes

At the end of the course the student will be able to

CO1	Design and implement user-friendly interfaces, handle data storage and integration effectively.
CO2	Implement DevOps tools and processes in the software development lifecycle.
CO3	Actively participate and collaborate within their project teams, demonstrating effective communication, teamwork, and problem-solving skills.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

CO-PO-PSO mapping

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2
CO 1			3		3								3	3
CO 2			3		3								3	3
CO 3								3	3	3				

Assessment Plan (for 50 marks of CIE)

Tool	Remarks	Marks
Internals	---	---
QUIZ	---	---
Lab Component	---	50
Alternate Assessment Tool	----	--
Total		50

Rubrics for Project Evaluation:

Rubrics	Full Stack Application Development Evaluation Rubric (20 Marks)				Total (50 marks)
Frontend Development (8 marks)	(8 marks) Professionally designed UI with intuitive user experience, responsive design, and attention to	(5 marks) Well-designed UI with responsive layout and user-friendly interactions.	(3 marks) Basic user interface (UI) components and layout implemented.	(0 marks) No or minimal frontend development implemented.	_ / 8



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

	detail.				
Backend Development (8 marks)	(8 marks) Robust backend implementation with efficient data management, authentication, and security measures.	(5 marks) Backend functionality implemented with proper data handling and integration.	(3 marks) Basic server-side logic with limited functionality.	(0 marks) No or minimal backend development implemented.	_ / 8
Data Storage and Integration (4 marks)	(4 marks) Advanced integration with multiple data sources and effective data management.	(3 marks) Integration with database or external services for data storage and retrieval.	(1 mark) Basic data storage implemented without integration.	(0 marks) No or minimal data storage or integration.	_ / 4
	DevOps Process Evaluation Rubric (15 Marks)				
Version Control (5 marks)	(5 marks) Advanced use of version control with clear commit messages, proper code review, and effective collaboration.	(4 marks) Effective use of version control with regular commits, proper branching, and merging strategies.	(3 marks) Basic version control implemented with some commits and branches.	(0 marks) Little or no use of version control.	_ / 5



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Continuous Integration (5 marks)	(5 marks) Advanced continuous integration setup with automated testing, code quality checks, and immediate feedback on failures.	(4 marks) Consistent implementation of continuous integration with automated builds and test execution.	(3 marks) Basic setup for continuous integration, but builds may fail occasionally.	(0 marks) No implementation of continuous integration.	_ / 5
Deployment Automation (5 marks)	(5 marks) Advanced automated deployment with zero-downtime deployment, canary releases, and blue/green deployment.	(4 marks) Automated deployment for multiple environments with rollback capability.	(3 marks) Basic automated deployment for specific environments.	(0 marks) Manual deployment process.	_ / 5
	Group Participation (5 marks)				
Group Participation (5 marks)	(5 marks) Active engagement, strong collaboration, and effective teamwork demonstrated throughout the project.	(4 marks) Regular participation, constructive contributions, and collaboration within the group.	(2 marks) Minimal participation with occasional contribution to group activities.	(0 marks) Lack of active participation and collaboration within the group.	_ / 5
	Presentation & Documentation (10) Marks				
Presentation (5 marks)	(6 marks) Professional presentation with	(4 marks) Well-structured presentation with clear	(2 marks) Basic presentation with limited	(0 marks) Incoherent and disorganized presentation	_ / 5



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

	engaging delivery, effective visuals, and comprehensive content.	content and adequate delivery.	structure and inconsistent delivery.	with inadequate content.	
Report & Documentation (5 marks)	(6 marks) Comprehensive and well-documented report covering all aspects of the project, including methodology, design, implementation, and future scope.	(4 marks) Well-structured report/documentation covering project details and implementation.	(2 marks) Basic report/documentation with limited content and organization.	(0 marks) Poorly structured and incomplete report/documentation.	_ / 5
TOTAL					_ / 50

Tutorial Plan (if applicable)

Laboratory Plan (if applicable)

Project Topics for Website Development:

Department Lab Stock Book Maintenance System; Department Faculty Weekly Report Submission System; Department Faculty Self-Assessment Report Submission System; Department Faculty Self –Appraisal form Submission System; Department Student Project Submission System; Department Conference Paper Submission System; College TEQIP student project proposal submission system; College TEQIP Faculty Workshop/Conference/Seminar Application Submission System; College Exam Application Form Submission System

Note: Apart from the above-mentioned project topics if student groups come up with any innovative project ideas which are useful for the Department / College academic purpose will be considered based on the approval and acceptance from class teacher.

Sl. No	Week	Activity	Content deliverables by the assigned teacher
1	1st	Formation of groups. Note: Student groups of size 2 or 3 or 4	Introduction to full stack web development



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

2	2nd	Project topic selection by each Group. Presentation: Student and Project topic introduction by each group	Version control using git , Github. Create basic web applications with back-end and database connection.
3	3rd	Design Layout of the Web Pages	Development and deployment of Web application.
4	4th ,5th , and 6th	Front end and back-end implementation	Docker
5	7th	Devops CI/CD pipeline	Docker Hub, CI/CD pipeline
6	8th and 9th	Design and Development of connecting among different web pages	
7	10th	Presentation by each group	
8	11th	Complete Project Work Demonstration by each group	
	12th	Project Report Preparation	

Text Book :

Supplementary texts and resources

1. A Beginners guide to DevOps

Basics: <https://www.edureka.co/blog/ebook/devops-ebook>

2. Learning DevOps: A <https://online-pmo.com/wp-content/Education/Learning%20DevOps.pdf>

Tutorial Link:

1. <https://www.coursera.org/learn/intro-to-devops>
2. <https://www.udacity.com/course/intro-to-devops--ud611>
3. <https://www.classcentral.com/subject/devops>

SEE Exam (50 Marks)

Evaluation of Projects carried out by students from the External examiner along with internal faculty.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Sem	IV		
Course Title	Linear Algebra and Optimization	Course Code	23MA4BSLAO
Credits	03	L – T – P	2-1-0
Contact hours	40		

Course Objectives:

The objectives of the course are to facilitate the learners to

- Appreciate the importance of linear algebra in computer and allied engineering science.
- Gain the knowledge of linear algebra tools and concepts to implement them in their core domain.
- Improve their mathematical thinking and acquire skills required for sustained lifelong learning.

Teaching-Learning Process (General Instructions)

These are sample Strategies; that teachers can use to accelerate the attainment of the various course outcomes.

- Lecture method(L) does not mean only traditional lecture method, but different type of teaching methods may be adopted to develop the outcomes.
- Encourage collaborative (Group Learning) Learning in the class.
- Ask HOT (Higher order Thinking) questions in the class, which promotes critical thinking.
- Adopt Problem-Based Learning (PBL), which fosters students' Analytical skills, and develops thinking skills such as the ability to evaluate, generalize, and analyze information rather than simply recall it.
- Discuss how every concept can be applied to the real world and when that's possible, it helps to improve the students' understanding.

UNIT-1

CONTINUOUS OPTIMIZATION – 1

[7 hours]

Function of several variables, partial differentiation, local and global optima, convex sets and functions separating hyperplanes, application of Hessian matrix in optimization, gradients of vector-valued functions, gradients of matrices, useful identities for computing gradients.

Teaching-Learning Process:	Chalk and Board, Problem-based learning
-----------------------------------	---

UNIT-2

CONTINUOUS OPTIMIZATION-2

[7 hours]

Optimization using gradient descent/ascent and NR method.

Sequential search 3-point search and Fibonacci search.

Constrained Optimization, Method of Lagrange multipliers, KKT optimality conditions.

Teaching-Learning Process:	Chalk and Board, Problem-based learning
-----------------------------------	---



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

UNIT-3

INNER PRODUCT SPACES

[7 hours]

Inner products, inner product spaces, length and orthogonality, orthogonal sets and Bases, projections, Gram-Schmidt orthogonalization process, QR-factorization, least squares problem and least square error. Curve fitting – Principle of least squares, fitting a straight line and fitting a parabola.

Teaching-Learning Process:	Chalk and Board, Problem-based learning
-----------------------------------	---

UNIT-4

EIGENVALUES AND EIGENVECTORS

[7 hours]

Introduction, Polynomials of Matrices, Cayley-Hamilton Theorem, eigen spaces of a linear transformation, Characteristic and Minimal Polynomials of Block Matrices, Jordan Canonical form.

Teaching-Learning Process:	Chalk and Board, Problem-based learning
-----------------------------------	---

UNIT-5

MATRIX DECOMPOSITION AND THEIR APPLICATIONS

[8 hours]

Diagonalization, Orthogonal diagonalization of real symmetric matrices, quadratic forms and its classifications, rank and signature of real quadratic forms, Singular value decomposition. Dimensional reduction – PCA.

Teaching-Learning Process:	Chalk and Board, Problem based learning
-----------------------------------	---

Course outcomes (Course Skills Set)

After successfully completing the course, the student will be able to understand the topics:

Course Code	CO	COURSE OUTCOME (CO)	PO	Strength
23MA4BSLIA	CO 1	Apply the concepts of linear algebra in Computer and Allied Engineering Sciences.	1	3
	CO 2	Demonstrate the applications of computer science and Allied Engineering Science using modern ICT tools.	1 & 5	3

Assessment Details (both CIE and SEE)

Component	Type of assessment	Max. Marks	Total	50 % Weightage	Total
CIE – Theory	Quiz	10	100	5	50
	AAT	10		5	
	Test 1	40		20	
	Test 2	40		20	
SEE	End Exam	100		50	

CIE methods/question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19
Autonomous Institute, Affiliated to VTU
DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

SEMESTER END EXAMINATION:

- Each unit consists of one full question.
- Five full questions to be answered.
- To set one question each from Units 1, 2 and 5 and two questions each from Units 3 and 4.

SUGGESTED LEARNING RESOURCES:

Text Books:

1. Linear Algebra and its applications, David C. Lay, Steven R. Lay, Judi J Mc. Donald, 6th Edition, 2021, Pearson Education.
2. Linear Algebra and its Applications, Gilbert Strang, 4th edition, 2005, Brooks Cole.
3. Linear Algebra: An Introduction, Richard Bronson & Gabriel B. Costa, 2nd edition, Academic press.

Reference Books:

1. Schaum's outline series -Theory and problems of linear algebra, Seymour Lipschutz, Marc Lipson, 6th edition, 2017, McGraw-Hill Education.
2. Linear Algebra and Optimization for Machine Learning, Charu C. Aggarwal, Springer, 2020
3. Linear Algebra, Stephen H. Friedberg, Arnold J. Insel and Lawrence E. Spence, Pearson, 2019, Fifth Edition.
4. Mathematics for Machine learning, Marc Peter Deisenroth, A. Aldo Faisal, Cheng Soon Ong, 2020, Cambridge University Press.
5. Linear Algebra, Kenneth Hoffman, Ray Kunze, 2nd edition, Pearson.

E-books and online course materials:

1. <https://ocw.mit.edu/courses/mathematics/18-06sc-linear-algebra-fall-2011/index.htm>
2. <https://www.math.ucdavis.edu/~linear/linear.pdf>

Online Courses and Video Lectures:

1. <https://www.coursera.org/learn/linear-algebra-machine-learning>
2. <https://nptel.ac.in/syllabus/111106051/>



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Sem	IV		
Course Title:	Theory of Computation		
Course Code:	23DC4ESTOC	Total Contact Hours: 40 hours	
L-T-P:	3-0-0	Total Credits:	3

Unit No.	Topics	Hours
1	Introduction to Finite Automata: Central Concepts of Automata Theory, Deterministic Finite Automata (DFA), Nondeterministic Finite Automata (NFA), Finite Automata with Epsilon Transition, An Application Text Search. NP Problems solvable in Polynomial Time, Satisfiability Problem	8
2	Regular Expressions and Languages: Regular Expressions, Finite Automata and Regular Expressions, Applications of Regular Expressions, Proving Languages Not to Be Regular, Closure Properties of Regular Languages, Equivalence and Minimization of Automata	8
3	Context Free Grammars and Languages Parse Trees: Context Free Grammars, Parse trees, Applications of Context Free Grammars, Ambiguity in Grammars and Languages, Eliminating Useless Symbols, Computing the Generating and Reachable Symbols, Eliminating Epsilon Productions, Eliminating Unit Productions, Chomsky Normal Form, Greibach Normal form	8
4	Pushdown Automata: Definition of the Pushdown Automaton, The Languages of a PDA, Equivalence of PDA's and CFG's, Deterministic Pushdown Automata, The Pumping Lemma for Context Free Languages, Closure Properties of Context Free Languages	8
5	Introduction to Turing Machine: Problems That Computers Cannot Solve, The Turing Machine, Programming Techniques for Turing Machines, Extensions to the Basic Turing Machine, Restricted Turing Machines, Turing Machines and Computers, Definition of Post Correspondence Problem, A Language That Is Not Recursively Enumerable, An Undecidable Problem That is RE, Other Undecidable Problems	8



B. M. S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Prescribed Text Book					
Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Introduction to Automata Theory, Languages and Computation	John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman: education	3 rd Edition	Pearson	2007

Reference Text Book					
Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Introduction to Languages and Automata Theory	John C Martin	3 rd Edition	Tata McGraw-Hill	2007
2.	An Introduction to formal Languages and Automata	Peter Linz	5 th Edition	Narosa Publishing House	2012

E-Book						
Sl. No.	Book Title	Authors	Edition	Publisher	Year	URL
1.	Introduction to Theory of Computation	Anil Mahe shwar i, Michi el Smid	-	Carleton Universit y	2019	https://cglab.ca/~michiel/TheoryOfComputation/TheoryOfComputation.pdf

MOOC Course				
Sl. No.	Course name	Course Offered By	Year	URL
1.	Automata Theory	edX	2022	https://www.edx.org/course/automata-theory
2.	Introduction to Automata, Languages and Computation	IITB	2022	https://onlinecourses.nptel.ac.in/noc21_cs19/preview
3.	Automata Theory	Stanford University	2022	https://online.stanford.edu/courses/soe-yicsautomata-automata-theory

Course Outcomes



B. M. S. COLLEGE OF ENGINEERING, BENGALURU-19

At the end of the course the student will be able to

CO1	Apply the knowledge of Automata Theory, Grammars & Regular Expressions for the given requirement of the formal language.
CO2	Analyze the given Automata to identify the formal language it represents.
CO3	Design Automata and Grammar for pattern recognition and syntax checking of the given formal language.

CO-PO mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12
CO 1	3											
CO 2		2										
CO 3			2									

Proposed Assessment Plan (for 50 marks of CIE)

Assessment Tool	No. of Assessments	Marks
Internals	2	40
QUIZ/AAT	2	10
Total		50

SEE Exam Question paper format

Unit-1	Mandatory	One Question to be asked for 20 Marks
Unit-2	Internal Choice	Two Questions to be asked for 20 Marks each
Unit-3	Internal Choice	Two Questions to be asked for 20 Marks each
Unit-4	Mandatory	One Question to be asked for 20 Marks
Unit-5	Mandatory	One Question to be asked for 20 Marks



B. M.S. COLLEGE OF ENGINEERING, BENGALURU-19
Autonomous Institute, Affiliated to VTU
DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Semester	IV		
Course Title:	Operating Systems		
Course Code:	23DC4PCOPS	Total Contact Hours: 40 hours	
L-T-P:	3-0-0	Total Credits:	3

Unit No.	Topics	Hours
1	Introduction to Operating Systems: What operating systems do, Operating System operations, Process management, Memory management, Storage management, Protection and security System Structures: Operating System Services, System calls, Operating System design and implementation, Operating System structure, System Boot.	8
2	Processes: Process Concept, Process Scheduling, Operations on Processes, Inter-process Communication. Threads: Overview, Multi-core Programming, Multithreading Models, Implicit Threading, Threading Issues. Process Synchronization- Background, The Critical section problem, Synchronization hardware, Mutex Locks, Semaphores, Classical problems of synchronization.	8
3	CPU Scheduling- Basic concepts, Scheduling criteria, Scheduling algorithms, Multiple-Processor scheduling. Deadlocks: System Model, Deadlock characterization, Methods for handling deadlocks, Deadlock prevention, Deadlock avoidance, Deadlock detection and recovery from deadlock.	8
4	Memory Management Strategies: Background, Swapping, Contiguous memory allocation, Paging, Structure of page table, Segmentation. Virtual Memory Management- Background, Demand paging, Page replacement, Thrashing.	8
5	Virtual Machines: Overview, Benefits and features, Building Blocks, Types of Virtual Machines and their implementations, Virtualization and Operating System Components, Protection Rings	8



B. M. S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

	Case Study: VMWare	
--	--------------------	--

Prescribed Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Operating System Concepts	Abraham Silberschatz, Peter Baer Galvin , Greg Gagne	9th Edition	John Wiley & Sons	2018
2.	Modern operating systems	Andrew Tanenbaum	4th Edition	Pearson Education	2009

Reference Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Operating System: Internals and Design Principles	William Stallings	8th Edition	Prentice Hall	2014
2.	Schaum's Outline of Operating Systems	J. Archer Harris	Kindle Edition	McGraw-Hill	2001

E-Book

Sl. No	Book Title	Authors	Edition	Publisher	Year	URL
1.	Operating Systems Course Notes	Dr. John T. Bell	-	University of Illinois Chicago	2006 & 2013	https://www.cs.uic.edu/~jbell/CourseNotes/OperatingSystems/index.html
2.	Operating System Concept	Abraham Silberschatz, Peter Baer Galvin , Greg	9th Edition	John Wiley & Sons	2018	https://drive.uqu.edu.sa/~mskhayat/files/MySubjects/2017SS%20Operating%20Systems/Abraham%20Sil



B. M.S. COLLEGE OF ENGINEERING, BENGALURU-19
Autonomous Institute, Affiliated to VTU
DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Proposed Assessment Plan (for 50 marks of CIE)

Assessment Tool	No. of Assessments	Marks
Internals	2	40
QUIZ/AAT	2	10
Total		50

SEE Exam Question paper format

Unit-1	Mandatory	One Question to be asked for 20 Marks
Unit-2	Internal Choice	Two Questions to be asked for 20 Marks each
Unit-3	Internal Choice	Two Questions to be asked for 20 Marks each
Unit-4	Mandatory	One Question to be asked for 20 Marks
Unit-5	Mandatory	One Question to be asked for 20 Marks



B. M. S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Semester	IV		
Course Title:	Design and Analysis of Algorithms		
Course Code:	23DC4PCDAA	Total Contact Hours: 40 hours	
L-T-P:	3-0-1	Total Credits:	4

Unit No.	Topics	Hours
1	Introduction to Algorithm, Fundamentals of Algorithmic Problem Solving. Analysis of Algorithm Efficiency: The Analysis Framework, Asymptotic Notations and Basic Efficiency Classes, Mathematical Analysis of Non Recursive Algorithm, Mathematical Analysis of Recursive Algorithms.	8
2	Brute-Force: String Matching, Exhaustive Search: TSP, Knapsack Problem, Assignment Problem, Depth-First Search and Breadth-First Search. Decrease-and-Conquer: Topological Sorting, Algorithms for Generating Combinatorial Objects: Generating Permutations, Decrease by-a-Constant-Factor Algorithms: Binary Search, Russian Peasant Multiplication, Variable Size-Decrease Algorithms: Computing Median and the Selection Problem	8
3	Divide-and-Conquer: Merge sort, Quicksort, Multiplication of Large Integers and Strassen's Matrix Multiplication. Transform-and-Conquer: Presorting, Heaps and Heap sort, Horner's Rule. Space and Time Tradeoffs: Horspool Algorithm, Boyer-Moore Algorithm.	8
4	Dynamic Programming: Coin Problem, The Knapsack Problem, Warshall's and Floyd's Algorithms. Greedy Technique: Prim's Algorithm, Kruskal's Algorithm-Without disjoint subsets and Union Find algorithms, Dijkstra's Algorithm, Huffman Trees.	8
5	Backtracking: n -Queens Problem, Subset-Sum Problem. Branch-and-Bound: Knapsack Problem, Traveling Salesman Problem. NP-Completeness: Polynomial time, Polynomial-time verification, NP-completeness and reducibility. NP-Complete Problems: The Clique	8



B. M. S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

	problem, The Vertex Cover problem, Approximation Algorithms: The Vertex-Cover problem.	
--	--	--

Prescribed Text Book					
Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Introduction to the Design and Analysis of Algorithms	Anany Levitin	Third Edition	Pearson	2014
2.	Introduction to Algorithms	Thomas H Cormen, Charles E Leiserson, Ronald L Rivest, Clifford Stein	Third Edition	The MIT Press	2009

Reference Text Book					
Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Fundamentals of Computer Algorithms	Ellis Horowitz, Satraj Sahni and Rajasekharam	2nd Edition	University Press Pvt. Ltd,	2009
2.	Analysis and design of Algorithms	Padma Reddy		Sri Nandi Publications	2009

E-Books						
Sl. No.	Book Title	Authors	Edition	Publisher	Year	URL
1.	Introduction to Design & Analysis of Algorithms	K. Raghava Rao	-	Smash words	2013	https://www.smashwords.com/books/view/365630



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

2.	Data structures and Algorithm Analysis in C++	Allen Weiss	Fourth edition	Pearson education	2014	http://www.uoitc.edu.in/images/documents/informatics-institute/Competitive_exam/DataStructures.pdf
----	---	-------------	----------------	-------------------	------	---

MOOC Courses

Sl. No.	Course name	Course Offered By	Year	URL
1	Algorithms	Coursera	2023	https://www.coursera.org/course/algs4partI
2	Design and Analysis of Algorithms	NPTEL	2023	https://onlinecourses.nptel.ac.in/noc19_cs47/preview

Course Outcomes

At the end of the course the student will be able to

CO1	Apply algorithmic design paradigms to basic computing problems.
CO2	Analyze the time complexity of different algorithms.
CO3	Design efficient algorithms using appropriate algorithm design techniques.
CO4	Conduct experiments to implement algorithms and provide valid conclusions.

CO-PO mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2		3										
CO3			3									
CO4				3	1							



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Proposed Assessment Plan (for 50 marks of CIE)

Assessment Tool	No. of Assessments	Marks
Internals	2	20
QUIZ/AAT	1	5
Lab Component	CIE+ Two Lab Tests	25
Total		50

Laboratory Plan

Instructions to Students to be followed in each lab:

1. Each Student should write down the program in the observation book and get it evaluated by the respective lab faculty in-charge and then execute the program.
2. Each Student should bring the lab record with the programs and output written for the programs completed in their respective previous week and get it evaluated by the lab faculty in-charge. In the record book students should - Handwrite the Program - Pasting of the printout of the Output or Handwriting of the Output (Output should be written for all the cases).
3. Students have to practice following list of programs and additional programming exercises will also be given in lab. Students will be made to solve coding challenges on platforms like LeetCode and HackerRank.

Lab Program	Unit#	Program Details
1	2	Write program to do the following: a. Print all the nodes reachable from a given starting node in a digraph using BFS method. b. Check whether a given graph is connected or not using DFS method.
2	2	Write program to obtain the Topological ordering of vertices in a given digraph.
3	2	Implement Johnson Trotter algorithm to generate permutations
4	3	Sort a given set of N integer elements using Merge Sort technique and compute its time taken. Run the program for different values of N and



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

		analyze its time complexity.
5	3	Sort a given set of N integer elements using Quick Sort technique and compute its time complexity.
6	3	Sort a given set of N integer elements using Heap Sort technique and analyze its time complexity.
7	4	Implement 0/1 Knapsack problem using dynamic programming.
8	4	Implement All Pair Shortest paths problem using Floyd's algorithm.
9	4	Find Minimum Cost Spanning Tree of a given undirected graph using Prim/Kruskal's algorithm.
10	4	From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm.
11	5	Implement "N-Queens Problem" using Backtracking.

SEE Question paper format

Unit-1	Mandatory	One Question to be asked for 20 Marks
Unit-2	Mandatory	One Question to be asked for 20 Marks
Unit-3	Internal Choice	Two Questions to be asked for 20 Marks each
Unit-4	Internal Choice	Two Questions to be asked for 20 Marks each
Unit-5	Mandatory	One Question to be asked for 20 Marks



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Sem	IV		
Course Title:	Internet of Things		
Course Code:	23IC4PCIOT	Total Contact Hours: 40 hours	
L-T-P:	3-0-0	Total Credits:	3

Unit No.	Topics	Hrs
1	IoT Network Architecture- Drivers Behind New Network Architectures, M2M: Machine-to-Machine communication, The oneM2M IoT Standardized Architecture, IoTWF Standardized Architecture, A simplified IoT Architecture, Difference between IoT and M2M, The core IoT Functional Stack, IoT Data Management and Compute Stack, Wireless sensor network in IoT: Introduction, Characteristics of Wireless Sensor Network, Types of WSN and their architecture.	8 Hrs
2	IoT Protocols- Communication Protocols: Serial Communication - UART, SPI, and I2C Protocols, Implementing Serial Communication with ESP 32-examples, Wireless Communication - Introduction to Wireless Modules -Wi-Fi, Bluetooth, BLE, Physical/Link Layer: IEEE 802.15.4 standard and ZigBee protocol, Low-Power Wide Area Networks (LPWAN)-LoRaWAN, NB-IoT, LTE	8 Hrs
3	IoT Protocols and IoT Platforms -IP as the IoT Network Layer: The need for Optimization, Optimizing IP for IoT-From 6LoWPAN to 6Lo, 6TiSCH, RPL, Transport Layer, Application Layer: CoAP, MQTT, Intel IoTivity – Device discovery functionality	8 Hrs
4	IoT Cloud services and Infrastructure- Amazon Web Services for IoT, Amazon EC2, Amazon AutoScaling, Amazon S3, Amazon RDS, Amazon DynamoDB, Amazon Kinesis, Amazon SQS, Amazon EMR, SDN and NFV for IoT: Software-defined Networking, Network Function Virtualization	8 Hrs
5	IoT Security-Security concepts in context to IoT Devices IoT Security Challenges, IoT System Security Domains, Security Issues in IoT Systems and Privacy Preservation, IoT Security Requirements, Security Technologies, IoT System Security Controls, Best Practices for Securing IoT Devices	8 Hrs



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Prescribed Text Books				
Book Title	Authors	Edition	Publisher	Year
IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things	David Hanes, Gonzalo Salgueiro		CISCO Press	2019
Internet Of Things - A Hands-On Approach	Arsheep Bahga, Vijay Madiseti		Universities Press	2015
Internet of Things	Surya Durbha, Jyoti Joglekar		OXFORD University Press	2021

ReferenceText Books				
Book Title	Authors	Edition	Publisher	Year
Introduction to IoT	Sudip Misra, Anandarup Mukherjee, Arijit Roy,		Cambridge University Press	2021
Internet of Things: Architecture and Design Principles	Raj Kamal	1st Edition	McGraw Hill Education	2017
The Internet of Risky Things	Sean Smith	1st Edition	O'Reilly Media	2017
Practical Internet of Things Security: Design a security framework for an Internet connected ecosystem	Brian Russell, Drew Van Duren	2nd Edition		2018

Course Outcomes

CO1	Understand the fundamental protocols, cloud and security issues in the Internet of things.
CO2	Apply various protocols, device discovery and cloud services in resource constraint networks for IoT applications
CO3	Analyze the various IoT architectural components.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

CO PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											
CO2		3										
CO3			3									

Tool	Remarks	Marks
Internals	Best 2 of 3	40
QUIZ		10
Total		50

MOOC Course				
Sl. No.	Course name	Course Offered By	Year	URL
1.	Introduction to Internet of Things	IIT Kharagpur	2018	https://nptel.ac.in/courses/106/105/106105166/
2.	AWS IoT: Developing and Deploying an Internet of Things	Edx	2020	https://www.edx.org/course/aws-iot-developing- and-deploying-an-internet-of-th

SEE Exam Question paper format

Unit-1	Mandatory	One Question to be asked for 20 Marks
Unit-2	Internal Choice	Two Questions to be asked for 20 Marks each
Unit-3	Internal Choice	Two Questions to be asked for 20 Marks each
Unit-4	Mandatory	One Question to be asked for 20 Marks
Unit-5	Mandatory	One Question to be asked for 20 Marks



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Semester	IV		
Course Title:	Computer Networks		
Course Code:	23IC4PCCNW	Total Contact Hours: 40 hours	
L-T-P:	3-0-1	Total Credits:	4

Unit No.	Topics	Hours
1	Introduction: Data Communications, Networks, Network Types, Network Models: Protocol Layering, Reference Models: The OSI Reference Model, The TCP/IP Reference Model, Physical Layer: Data and signals Digital Transmission, (D-D Conversion) Bandwidth Utilization, Multiplexing, Switching, Circuit Switched Networks, Packet Switching.	8
2	Data Link Layer: Link Layer Addressing, Error Detection and Correction, Block Coding, Cyclic Codes, Checksum. Data Link Control: DLC Services, Data-Link Layer Protocols, Media Access Control.	8
3	Network Layer: Network Layer Services, Packet Switching, Network Layer Performance, IPV4 Addresses. Network Layer Protocols: Internet Protocol, ICMPV4, Unicast Routing, Routing algorithms, Unicast routing protocols, Internet Structure, Routing Information Protocol (RIP), Next Generation IP: IPV6 Addressing, IPV6 Protocol, ICMPv6 Protocol, Transition from IPV4 to IPV6.	8
4	Transport Layer: Transport Layer Protocols, User Datagram Protocol, Transmission Control Protocol.	8
5	Application Layer: Introduction, Standard Client Server Protocols, DNS—The Internet's Directory Service, SMTP, SNMP, FTP	8

Prescribed Text Book					
Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Data Communications and Networking	Behrouz A Forouzan	5 th Edition	McGraw Hill	2013
2.	Computer Networks	Andrew S. Tanenbaum, David J. Wetherall	5th Edition	Pearson	2011
Reference Text Book					
Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Data and Computer Communication	William Stallings	8th Edition	Pearson Education	2008
2.	Computer Networks – A Systems Approach	Larry L. Peterson and Bruce S. Davie	4th Edition	Elsevier	2007



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

E-Book						
Sl. No.	Book Title	Authors	Edition	Publisher	Year	URL
1.	An Introduction to Computer Networks	Peter L Dordal	1 st Edition	-	2020	https://intronetworks.cs.luc.edu/current/ComputerNetworks.pdf
2.	A Top-Down Approach: Computer Networking	James F Kurose & Keith W Ross	8 th Edition	Pearson	2021	https://gaia.cs.umass.edu/kurose_ross/online_lectures.htm

MOOC Course				
Sl. No.	Course name	Course Offered By	Year	URL
1.	Computer Networking	Coursera	2023	https://www.coursera.org/learn/illinois-tech-computer-networking
2.	NOC: Computer Networks and Internet Protocol	NPTEL		https://nptel.ac.in/courses/106105183

Course Outcomes

At the end of the course, the student will be able to

CO1	Apply the fundamental concepts of communication in networking.
CO2	Analyze the various protocols, and techniques in TCP/IP network architecture
CO3	Develop applications that demonstrate the functionalities of physical, Data Link, Network, Transport or Application layer

CO-PO mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2		3										
CO3			3		1							

Proposed Assessment Plan (for 50 marks of CIE)

Assessment Tool	No. of Assessments	Marks
Internals	2	40
QUIZ/AAT	2	10
Total		50



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19
Autonomous Institute, Affiliated to VTU
DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

SEE Exam Question paper format

Unit-1	Mandatory	One Question to be asked for 20 Marks
Unit-2	Internal Choice	Two Question to be asked for 20 Marks each
Unit-3	Internal Choice	Two Question to be asked for 20 Marks each
Unit-4	Mandatory	One Question to be asked for 20 Marks
Unit-5	Mandatory	One Question to be asked for 20 Marks



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19
Autonomous Institute, Affiliated to VTU
DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

COMPUTER NETWORKS Lab - Plan of Activities
CYCLE 1: Exercises done using CISCO Packet Tracer
CYCLE 2: Execution of Lab Programs using C/C++/Python

CYCLE 1: Exercises done using CISCO Packet Tracer			
Experiment	Unit	Name of Experiment	Remarks
1	2	Create a topology and simulate sending a simple PDU from source to destination using hub and switch as connecting devices and demonstrate ping message.	
2	3	Configure IP address to routers in packet tracer. Explore the following messages: ping responses, destination unreachable, request timed out, reply	
3	3	Configure default route, static route to the Router	
4	5	Configure DHCP within a LAN and outside LAN.	
5	3	Configure RIP routing Protocol in Routers	
6	3	Configure OSPF routing protocol	
7	3	Demonstrate the TTL/ Life of a Packet	
8	5	Configure Web Server, DNS within a LAN.	
9	2	To construct simple LAN and understand the concept and operation of Address Resolution Protocol (ARP)	
10	5	To understand the operation of TELNET by accessing the router in server room from a PC in IT office.	
11	3	To construct a VLAN and make the PCs communicate among a VLAN	
12		To construct a WLAN and make the nodes communicate wirelessly	
CYCLE 2: Execution of Lab Programs using C/C++/Python			
13	2	Write a program for error detecting code using CRC-CCITT (16-bits).	
14	3	Write a program for congestion control using Leaky	



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

		bucket algorithm.	
15	4	Using TCP/IP sockets, write a clientserver program to make client sending the file name and the server to send back the contents of the requested file if present.	
16	4	Using UDP sockets, write a clientserver program to make client sending the file name and the server to send back the contents of the requested file if present.	
17	3,4,5	Tool Exploration -Wireshark	



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Semester	4		
Course Title:	Hardware Interface Design		
Course Code:	23IC4AEHID		
L-T-P:	0-0-1	Total Credits:	1

A Introduction:

Familiarization with the concept following concepts of IOT for Arduino / Raspberry Pi and necessary software installation.

- Bluetooth (HC05) and LED with ESP32. Write interfacing to Control LED (On/Off) using Bluetooth Serial Controller App.
- Design an Interface circuit for Ultrasonic Sensor (HC-SR04) with ESP32. Write a program to measure the distance and display it on an LCD.
- Design and implement a Fire alert system using GSM.
- Design an Interface for DHT11 (Temperature and Humidity) Sensor with ESP32. Write a Program to display temperature and humidity on the Blynk App.
- Design an Interface for DHT11 (Temperature and Humidity) Sensor with ESP32. Write a Program to display temperature and humidity using the FireBase server.
- Program to upload and retrieve temperature and humidity data from thingspeak cloud platform.
- To install MySQL database for any board and perform basic SQL queries.

B Course Outcomes:

At the end of the course the student will be able to

CO2	Design and develop a sustainable model
CO3	Integrate hardware interface to communicate with devices and cloud
CO4	Prepare budget and demonstration of project



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

c CO-PO-PSO mapping

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO 1	3				3								3	3
CO 2			3		3	3	3						3	3
CO 3			3		3								3	3
CO 4								3	3	3	2			

D Assessment Plan (for 50 marks of CIE)

Tool	Remarks	Marks
Internals	---	---
QUIZ	---	---
Lab Component	---	50
Alternate Assessment Tool	----	--
Total		50

E Rubrics for Project Evaluation:

Sl No	Criteria	Excellent	Good	Satisfactory
1	Creativity, Originality and Planning	(10 marks) Project is creative, original and well planned (3)	(8 marks) Project is moderately creative, original and planned (1.5)	(6 marks) Project is not creative, neither original nor planned (1)
2	Hardware circuit Design	(10 marks) All modules are designed appropriately in accordance with the	(8 marks) All modules are designed moderately in accordance with the requirements. (1.5)	(6 marks) Very basic design. All modules are designed inappropriately in



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

		requirements. (3)		accordance with the requirements. (1)
3	Implementation	(10 marks) Ideal implementation of modules for all set objectives. (4)	(8 marks) Appropriate implementation of modules for most of the set objectives. (2)	(6 marks) Concerned about implementation of modules for a few set objectives. (1)
4	Report	(8 marks) Comprehensive report covering all project aspects with meticulous documentation, including methodology, design, budget and future scope.	(6 marks) Well-structured report with detailed coverage of project implementation.	(4 marks) Basic report with limited content, covering essential project details.
5	Presentation	(8marks) Professional, engaging presentation with outstanding visuals and comprehensive content, demonstrating exceptional delivery skills.	(4 marks) Well-structured presentation with clear content and effective delivery.	(2 marks) Basic presentation with some structure and varying delivery quality.
6	Participation in Discussions	(4marks) Answer questions/queries professionally. Good delivery	(2marks) Participated in discussions; on some occasions (0.75)	(1marks) Listened mainly; Rarely spoke up,



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

F Tutorial Plan (if applicable)

G Laboratory Plan (if applicable)

Sl. No	Week	Activity	Content deliverables by the assigned teacher
1	1st	Formation of groups. Note: Student groups of size 2 or 3 or 4	Introduction to ESP32
2	2nd	Project topic selection by each Group. Presentation: Student and Project topic introduction by each group	Verify feasibility and give suggestions
3	3rd to 6th	Conduct experiments	Experiments on integrating the board to Cloud, GSM, Wifi and Bluetooth
4	7th to 9th	Design and implement the project	Guide and give suggestions
5	10th	Presentation by each group	Integrating Feedback & Refining Applications
6	11th	Complete Project Work Demonstration by each group	Final Project Presentations and Submissions
7	12th	Project Report Preparation	



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Semester	V		
Course Title:	Cyber Security		
Course Code:	23IC5PCCSY	Total Contact Hours:	40 hours
L-T-P:	3-0-1	Total Credits:	4

Unit No.	Topics	Hours
1	Introduction to Cyber Security: Basic Cyber Security Concepts, layers of security, Vulnerability, threat, Harmful acts, Internet Governance – Challenges and Constraints, Computer Criminals, CIA Triad, Assets and Threat, motive of attackers, active attacks, passive attacks, Software attacks, hardware attacks, Spectrum of attacks, Taxonomy of various attacks, IP spoofing, Methods of defense, Security Models, risk management, Cyber Threats- Cyber Warfare, Cyber Crime, Cyber terrorism, Cyber Espionage, etc., Comprehensive Cyber Security Policy.	8
2	Cybercrime: Mobile and Wireless Devices- Introduction, Proliferation of Mobile and Wireless Devices, Trends in Mobility, Credit Card Frauds in Mobile and Wireless Computing Era, Security Challenges Posed by Mobile Devices, Registry Settings for Mobile Devices, Authentication Service Security, Attacks on Mobile/Cell Phones, Mobile Devices: Security Implications for Organizations, Organizational Measures for Handling Mobile, Organizational Security Policies and Measures in Mobile Computing Era, Laptops.	8
3	Tools and Methods Used in Cybercrime: Introduction, Proxy Servers and Anonymizers, Phishing, Password Cracking, Keyloggers and Spywares, Virus and Worms, Trojan Horses and Backdoors, Steganography, DoS and DDoS Attacks, SQL Injection, Buffer Overflow, Attacks on Wireless Networks. Phishing and Identity Theft: Introduction, Phishing, Identity Theft (ID Theft).	8
4	Understanding Computer Forensics: Introduction, Historical Background of Cyber Forensics, Digital Forensics Science, The Need for Computer Forensics, Cyberforensics and Digital Evidence, Forensics Analysis of E-Mail, Digital Forensics Life Cycle, Chain of Custody Concept, Network Forensics, Approaching a Computer Forensics Investigation, Setting up a Computer Forensics Laboratory: Understanding the Requirements, Computer Forensics and Steganography, Relevance of the OSI 7 Layer Model to Computer Forensics, Forensics and Social Networking Sites: The Security/Privacy Threats, Computer Forensics from Compliance Perspective, Challenges in Computer Forensics, Special Tools and Techniques, Forensics Auditing, Anti Forensics.	8



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

5	Cybersecurity- Organizational Implications: Introduction, Cost of Cybercrimes and IPR issues, Web Threats for organizations, Security and Privacy implications from Cloud Computing, Social Media marketing, Social Computing and the Associated Challenges for Organizations, Organizational guidelines, Incident Handling, Cybersecurity Standards and Frameworks: DFARS, ISO 22301, HIPAA, NIST CSF Cybercrime and Cyberterrorism: Intellectual Property in the Cyberspace, The Ethical Dimension of Cybercrimes, The Psychology, Mindset and Skills of Hackers and Cybercriminal, Sociology of Cybercriminals, Information warfare.	8
---	---	---

Prescribed Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Cyber Security: Understanding Cyber Crimes, Computer Forensics And Legal Perspectives	Sunit Belapure and Nina Godbole	1st	Wiley India, .	2013
2.	Computer and Cyber Security: Principles, Algorithm, Applications, and Perspectives	B. B. Gupta, D. P. Agrawal, Haoxiang Wang	1st	CBS publishers & distributors Pvt. Ltd	2018

Reference Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Cyber SecurityEssentials,	James Graham, Richard Howard and Ryan Otson,	1st	CRC Press.	2010
2.	Introduction to Cyber Security	Chwan-Hwa(john) Wu,J.David Irwin.	1st	CRC Press.	2013

E-Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year	URL
1.	CyBOK: The Cyber Security Body of Knowledge	Awais Rashid,Howard Chivers ,George Danezis, Emil Lupu, Andrew Martin -	1st	University of Bristol	2019	https://www.cybok.org/knowledgebase/



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

2.	Cybersecurity: The Beginner's Guide	Dr. Erdal Ozkaya	1st	Packt Publishing Ltd.	2019	https://www.jre-training.com/WebSecurity/Cybersecurity.pdf
3.	CyberSecurity for beginners	Heimdall Security team	-	heimdalsec urity	2011	https://heimdalsecurity.com/pdf/cyber_security_for_beginners_ebook.pdf

MOOC Course

Sl. No.	Course name	Course Offered By	Year	URL
1.	Cyber Security	Swayam	2020	https://onlinecourses.swayam2.ac.in/cec20_cs15/preview
2.	Cyber Security and Privacy NPTEL	Swayam	2023	https://onlinecourses.nptel.ac.in/noc23_cs127/preview
3.	Introduction to Cyber Security	Swayam		https://onlinecourses.swayam2.ac.in/nou19_cs08/preview

Course Outcomes

At the end of the course, the student will be able to

CO1	Apply the concepts of cyber security, and cyber-crime.
CO2	Analyze threats and risks within the context of cyber security.
CO3	Design applications for cybercrime using appropriate cyber security methods.
CO4	Conduct experiments to implement cyber attacks and its countermeasures.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

CO-PO mapping

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3							2					2		
CO2		3													
CO3			3		3			2							3
CO4				2	3										3

Proposed Assessment Plan (for 50 marks of CIE)

Assessment Tool	No. of Assessments	Marks
Internals	2	20
QUIZ	1	05
Lab	CIE + 1 Test	25
Total		50

SEE Exam Question paper format

In all five units, two questions are to be asked for 20 marks each as internal choice.

Bloom's Level	Percentage of Questions to be Covered
Remember / Understand	40%
Apply / Analyse	40%
Create / Evaluate	20%

LAB EXPERIMENTS	
Experiment No.	Name of the Experiment
1	Installing and Configuring Kali Linux. Write a step-by-step guide to install Kali Linux on a virtual machine or a physical machine. This guide should cover the entire installation process, from downloading the ISO image to completing the installation and setting up the environment
2	Configure a firewall using iptables on a Linux system, write rules to filter traffic, and test its effectiveness against various network attacks.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

3	Perform a network vulnerability scan using tools like Nessus or OpenVAS. Analyze the scan results to identify and prioritize vulnerabilities for remediation.
4	Develop and configure a simple Intrusion Detection System using Snort/WireShark to monitor network traffic and detect potential security breaches.
5	Capture and analyze network traffic for a simulated Advanced Persistent Threat (APT). Reconstruct the attack timeline using network forensic tools like Wireshark, NetworkMiner, Snort.
6	Simulate a Distributed Denial of Service (DDoS) attack in a controlled environment using LOIC/Slowloris, and analyze its impact on network performance.
7	Develop a custom keylogger in Python to capture keyboard inputs, and discuss ethical implications and countermeasures to detect such tools.
8	Analyze a malware sample in a controlled environment using Cuckoo Sandbox/ WireShark, identify its behavior, and suggest mitigation strategies.
9	Write a script to automate the process of scanning for IoT devices on a network using Shodan, identify vulnerabilities, and suggest security enhancements.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Semester	V		
Course Title:	Machine Learning		
Course Code:	23IC5PCMLG	Total Contact Hours:	40 hours
L-T-P:	3-0-1	Total Credits:	4

Unit No.	Topics	Hours
1	Machine Learning Landscape: Introduction, Types of Machine Learning, Challenges of Machine Learning, Testing and Validating. Supervised Learning Decision Tree Learning: Decision tree representation, Appropriate problems for decision tree learning, Basic decision tree learning algorithm, Issues in Decision tree learning, CART Training algorithm	8
2	Support Vector Machines: Linear SVM, Nonlinear SVM, SVM Regression, Under the Hood. Instance Based Learning: Introduction, k-Nearest Neighbor learning	8
3	Probabilistic Learning Bayesian Learning: Bayes Theorem and Concept Learning, Maximum Likelihood, Minimum Description Length Principle, Bayes Optimal Classifier, Gibbs Algorithm, Naïve Bayes Classifier, Bayesian Belief Network, EM Algorithm.	8
4	Ensemble Learning and Random Forests: Voting Classifiers, Bagging and Pasting, Random Patches and Random Subspaces, Random Forests, Boosting, Stacking	8
5	Unsupervised Learning Techniques Clustering – Kmeans, DBSCAN, Other Clustering Algorithms, Gaussian Mixtures – Anomaly Detection, Selecting Clustering, Bayesian Gaussian Mixture Models, Other algorithms for anomaly and novelty detection Reinforcement Learning: Markov Decision Process, Introduction, Learning Task, Q Learning	8

Prescribed Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Machine Learning	Tom M. Mitchell	1st	McGraw Hill Education	2013
2	Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow	Aurelien Geron	2nd	O'Reilly	2020

Reference Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Introduction to Machine Learning	Andreas C Muller &	First	Shroff Publishers	2019

B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

	with Python	Sarah Guido			
2.	Thoughtful Machine learning	Mathew Kirk	First	Shroff Publishers	2019

MOOC Course

MOOC Course				
Sl. No	Course name	Course Offered By	Year	URL
1.	Machine Learning	Coursera	--	https://www.coursera.org/learn/machine-learning
2.	Introduction to Machine learning	NPTEL	2016	https://swayam.gov.in/nd_noc20_cs29/preview

E-Book

E-Book						
Sl. No.	Book Title	Authors	Edition	Publisher	Year	URL
1.	The Elements of Statistical Learning	Trevor Hastie, Robert Tibshirani, Jerome H. Friedman	Second	Springer	2009	https://web.stanford.edu/~hastie/Papers/ESLII.pdf
2.	Machine Learning in Action	Peter Harrington	First	Manning	2017	http://www2.ift.ulaval.ca/~chaib/IFT-4102-7025/public_html/Fichiers/Machine_Learning_in_Action.pdf

Course Outcomes

At the end of the course the student will be able to

CO1	Apply different learning algorithms for various complex problems
CO2	Analyze the learning techniques for given dataset
CO3	Design a model using machine learning to solve a problem.
CO4	Conduct practical experiments to solve problems using appropriate machine learning techniques.

CO-PO mapping

[illegible]



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19
Autonomous Institute, Affiliated to VTU
DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Proposed Assessment Plan (for 50 marks of CIE)

Tool	Remarks	Marks
Internals	2	25
QUIZ	1	5
Lab Component	CIE + 1 Lab Test	25
Total		50

SEE Exam Question paper format

In all five units, two questions are to be asked for 20 marks each as internal choice.

Bloom's Level	Percentage of Questions to be Covered
Remember / Understand	25%
Apply / Analyse	50%
Create / Evaluate	25%



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19
Autonomous Institute, Affiliated to VTU
DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Semester	V		
Course Title:	Cryptography		
Course Code:	23IC5PCCRP	Total Contact Hours: 40 hours	
L-T-P:	3-1-0	Total Credits:	4

Unit No.	Topics	Hours
1	Introduction: Security Goals, Cryptographic Attacks Mathematics of Cryptography: Integer Arithmetic, Modular Arithmetic, Linear Congruence Traditional symmetric-Key Ciphers: Introduction, Substitution Ciphers, Transposition Ciphers, Mathematics of Symmetric-key cryptography: Algebraic Structures, GF (2^{m-n}) Fields.	8
2	Introduction to Modern Symmetric Key Ciphers: Modern Block Ciphers, Modern Stream Ciphers. Data Encryption Standard (DES): Introduction, DES Structure, DES Analysis, Security of DES, Multiple DES. Advanced Encryption Standard (AES): Introduction, Transformations, Key Expansion, AES Ciphers, analysis of AES.	8
3	Encipherment using Modern Symmetric-Key Ciphers: Use of Modern Block Ciphers, Use of Stream Ciphers. Mathematics of Asymmetric-Key Cryptography: Primes, Primality Testing, Chinese Remainder Theorem, Quadratic Congruence, Legendre Symbol.	8
4	Asymmetric -Key Cryptography: Introduction, RSA cryptosystem, ElGamal Cryptosystem, Elliptic Curve cryptosystems. Cryptographic Hash functions: Introduction Secure hash algorithm, SHA-512	8

**B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19**

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

5	Message Integrity and Message Authentication: Message authentication, Digital Signature, RSA digital signature. Key Management: KERBEROS , Diffie-Hellman Key Agreement, X.509	8
---	--	---

Prescribed Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Cryptography and Network Security	Behrouz A. Forouzan and Debdeep Mukhopadhyay	2 nd	Tata McGraw Hill	2013
2.	Cryptography and Network Security Principles and practice	William Stallings	5th	Pearson Education Asia	2013

Reference Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Cryptography: Theory and Practice	Stinson. D.	3rd	Chapman & Hall/CRC	2012
2.	Cryptography and Network Security	Atul Kahate	3rd	Tata McGraw-Hill Publishing	2008

E-Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year	URL
1.	Cryptography and Network Security. Principles and Practice	William Stallings	3 rd	Pearson Education	2007	http://williamstallings.com/Crypto3e.html
2.	Handbook of Applied Cryptography	Menez, van Oorschot, Vanstone	ISBN: 0-8493-8523-7	CRC Press	2001	http://www.cacr.math.uwaterloo.ca/hac/

MOOC Course



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Sl. No.	Course name	Course Offered By	Year	URL
1.	Cryptography and Network Security	NPTEL	2022	https://onlinecourses.nptel.ac.in/noc22_cs90/preview
2.	Cryptography 1	Coursera	2019	https://www.coursera.org/course/crypto

Course Outcomes

At the end of the course the student will be able to

CO1	Apply cryptographic techniques to ensure data confidentiality, integrity, and authentication.
CO2	Analyze various symmetric and asymmetric cryptosystems and types of attacks on these cryptosystems.
CO3	Demonstrate cryptographic encryption and decryption techniques.

CO-PO mapping

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	3												2		
CO2		3													
CO3				2	3				1	1					3

Proposed Assessment Plan (for 50 marks of CIE)

Assessment Tool	No. of Assessments	Marks
Internals	2	40
AAT	1 (Usage of CrypTool)	10
Total		50

Proposed Tutorial Plan:

Tutorial	Topic
----------	-------



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

No.	
1	Introduction to CrypTool and Installation
2	Demonstration of basic features available in CrypTool
3	Demonstration of Caesar cipher In the message to decode, any punctuation is left unchanged in the encoded message, as too are any numbers. To change this Options > Text Options and from here you can select what attributes of a message the cipher will alter and which it will leave unchanged. Experiment encrypting the same message with the Caesar cipher with different settings selected from the text options. Decipher each message after doing so and see if the deciphered message still has the same punctuation, spacing etc.
4	Demonstration of Vigenere cipher Animal is a tool within the CrypTool that displays the concepts behind a cipher in a user-friendly fashion, by means of an animation. Demonstrate the use of animal tool for the above cipher.
5	Demonstration of DES Open a new file and type a plaintext message. Next click from the menu Crypt/Decrypt > Symmetric (modern) > DES (ECB)... This presents a key selection window, this key must be 64 bits long, which equates to 16 hexadecimal figures. For simplicity use the default key of: 00 00 00 00 00 00 00 00 Select Encrypt and there should be presented a window showing the data encrypted in hexadecimal form and its corresponding ASCII representation. To decrypt the message again select Crypt/Decrypt > Symmetric (modern) > DES (ECB)... Use the same key and select Decrypt , and the original message will be displayed in hexadecimal representation. Selecting View > Show as text displays it in ASCII; you may also notice some of the formatting is lost in the process or some padding is added. Encrypt the same message using the same process as above only selecting Crypt/Decrypt > Symmetric (modern) > DES (CBC)... instead. Compare the two encrypted messages.
6	Compare ECB versus CBC mode of operation for the following applications: • An online bank statement • An encrypted VoIP session • Viewing of a website using TCP/IP
7	Demonstrate DES encryption and decryption using Animal.
8	Demonstration of RSA Now, encrypt a message of your choice using the values: $p = 59, q = 71, e = 13$ Observe the results. Encrypt the same message with the values: $p = 673, q = 619, e = 13$
9	Demonstrate RSA encryption and decryption using Animal.
10	Demonstrate RSA implementation using PKI.
11	1963497163 is the product of two prime numbers, use tools within the CrypTool to find these two prime numbers. Mention what tools you used to do this.
12	Demonstrate hybrid encryption Combine aspects of AES and RSA algorithms and demonstrate encryption of different plaintext.
13	Use CrypTool to generate the plaintext for the following decoded message: KIRIAGANOTGFPUOPNUCDPOFIHLVQZDCOMWNKDOUPMIDFCRDFLZ BNPZZSNBNBQGZMIMCROCKCTITMDFAWWXAHCAGBCPASKRNB



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

AAT Plan:

Students are supposed to implement any of the encryption /decryption algorithms with change of various parameters and show the various possible attacks on the ciphers. Also, demonstration of how they can be overcome using suitable methods.

Sl. No	Week	Activity
1	1 st and 2 nd	Formation of groups. Note: Student groups of size 2 members only
2	3 rd	AAT topic selection by each group
3	4 th	Presentation: Student team and topic introduction by each group
4	5 th , 6 th	Design the workflow along with Front-end Design or Tool usage introduction
5	7 th	Presentation on Front-end Design of the application or Tool features demonstration
6	8 th , 9 th , 10 th	Design and Development of the actual algorithm and testing it for various test cases. Demonstration of the tool used and the task involved.
7	11 th	Complete code demonstration or complete demonstration of the task using the tool
8	12 th	AAT Report Preparation

Rubrics used for evaluation (AAT):

Criteria	Exemplary	Proficient	Partially Proficient	Points
User Interface / Front End Design OR Tool Usage	(1) The designed application has an exceptional design, attractive and usable interface. It is easy to locate all important elements.	(0.75) The designed application has an attractive design and usable interface. It is easy to locate all important elements.	(0.25) The designed application has a usable design interface, but may appear busy or boring. It is easy to locate most of the important elements.	____ / 1



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Implementati on of the Algorithm OR Implementati on done in the Tool	(4) Implementation of the algorithm has been done accurately without the usage of any library functions.	(2.5) Implementation of the algorithm has been done appropriately without the usage of any library functions.	(1.5) Implementation of the algorithm has been done with usage of a few library functions.	____ /4
Testing for various cases	(1) The implemented algorithm works for any given valid input.	(0.75) The implemented algorithm works for almost all valid inputs.	(0.5) The implemented algorithm works for any valid inputs.	____ /1
Application/R elevance	(1) The designed algorithm has several applications and is relevant in the area of cryptography.	(0.75) The designed algorithm has few applications and is relevant in the area of cryptography.	(0.5) The designed algorithm has few applications and is not very relevant in the area of cryptography.	____ /1
Report	(1) Clear and Effective writing and adherence to appropriate style guidelines	(0.75) Writing that is clear and effective for the most part and minor errors in adherence to appropriate style guidelines	(0.5) Unclear and ineffective writing and multiple errors in adherence to appropriate style guidelines	____/1
Oral communicati on (presentation)	(1) Clear and effective communication	(0.75) Communication is clear	(0.5) Unclear communication	____/1



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Participation in Discussions	(1) Provided many good ideas; inspired others; clearly communicated ideas, needs, and feelings.	(0.75) Participated in discussions; on some occasions, made suggestions.	(0.5) Listened mainly; Rarely spoke up, and ideas were off the mark.	___/1
Total				___/ 10

SEE Exam Question paper format

In all five units, two questions are to be asked for 20 marks each as internal choice.

Bloom's Level	Percentage of Questions to be Covered
Remember / Understand	35%
Apply / Analyse	40%
Create / Evaluate	25%



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Semester	V		
Course Title:	Unix System & Network Programming		
Course Code:	23IC5PCUNP	Total Contact Hours:	30 hours
L-T-P:	0-1-0	Total Credits:	1

Tutorial Plan:

Unit No.	Topics	Hours
1	UNIX Architecture and Command Usage: Unix Architecture, Features of UNIX, POSIX and the Single UNIX Specification, Internal and External Commands, Command Structure, Flexibility of Command Usage. • <i>Application of the file manipulation system calls for different types of files</i> The File System: The File, What's in a File name, The Parent-Child Relationship, the HOME variable, pwd, cd, mkdir, rmdir, Absolute Pathnames, Relative Pathnames, ls, The UNIX File System. 4. <i>Demonstrate copy of a file using system calls.</i> 5. <i>Output the contents of its Environment list</i>	6
2	Parent and Child Process: 4. <i>Demonstrate the implementation of a child process from the parent process.</i> 5. <i>Demonstrate a program to create zombie processes and orphan processes.</i>	6
3	Unix Socket - Client Server Model: Unix Socket - Structures, Unix Socket - Ports and Services, Unix Socket - Helper Functions. Echo Client & server: echo client, echo server, file descriptors • <i>Demonstrate an Echo Client and Echo server using TCP to estimate the round trip</i>	6
4	Remote Procedure Call: Introduction of RPC, RPC Call, Types, Features, Stub. • <i>Demonstrate a fully working remote procedure call (RPC) program</i> Advanced Socket System Calls: Programs to demonstrate the usage of Advanced socket system calls like getsockopt(), setsockopt(), getpeername(), getsockname(), readv() and writev(). 3. <i>Demonstrate program for Remote Command Execution using sockets</i>	6
5	Pipes and FIFO: Simple client server example, Pipes, Full- Duplex Pipes, FIFOs, Pipes and FIFO Limits 4. <i>Demonstrate a program for bi- directional communication using 2 pipes</i> I/O Multiplexing: I/O Model, Nonblocking I/O Model, I/O Multiplexing Model 4. <i>Demonstration a program for Non-Blocking I/O</i>	6

Prescribed Text Book



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	UNIX System Programming using C++	Terrance Chan	1st	Pearson Education	2008
2	Advanced Programming in the UNIX Environment	W. Richard Stevens	5th	Pearson Education	2001

Reference Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	UNIX & Shell Programming	M.G. Venkatesh Murthy	2nd	Pearson Education	2007
2.	The Complete Reference UNIX	Kenneth Rosen, Douglas Host, Rachel Klee, James Farber, Richard Rosinski	2nd	Tata McGRAW HILL Edition	2008

MOOC Course

Sl. No.	Course name	Course Offered By	Year	URL
1.	Linux Shell Scripting: A Project-Based Approach to Learning	Udemy	2020	https://www.udemy.com/course/linux-shell-scripting-projects/?couponCode=SKILLS4SALEA

Course Outcomes:

At the end of the course the student will be able to

CO1	Apply knowledge of UNIX Operating Systems and network programming protocol for different applications.
CO2	Analyze the client-server model using networking constructs.
CO3	Design and develop solutions for networking applications.

CO-PO-PSO mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3												3		
CO2		3													
CO3			3						1	1			3		



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Proposed Assessment Plan (for 50 marks of CIE)

Assessment Tool	No. of Assessments	Marks
Internals	1	40
Quiz	1	10
Lab Component	---	---
AAT	---	---
Total		50

SEE Exam (50 Marks)

SEE will be conducted through quiz for 50 marks.

Sem	V		
Course Title:	Research Methodology and IPR		
Course Code:	23DC5AERMI	Total Hours : 40 hours	
L-T-P:	3-0-0	Total Credits:	3

Unit No.	Topics	Hrs
1	Research Methodology: An Introduction - Meaning of Research, Objectives of Research, Types of Research, Research Approaches, Significance of Research, Research methods vs Methodology, Research and scientific method, Research Process, Criteria of Good Research. Define the Research Problem - What is research problem, Selecting the problem, Necessity of Defining the problem, Technique involved in Defining a Problem Research Design - Meaning of Research Design, Need for Research Design, Features of Good Design, Important concepts Relating to Research Design, Different Research Design.	8
2	Design of Sample Surveys - Introduction, Sample Design, Sampling and Non-Sampling Errors, Sample Survey vs Census Survey, Types of Sampling Designs Chi-Square Tests - Test of Difference of more than Two proportions, Test of Independence of Attributes, Test of Goodness of Fit Analysis of Variance - The ANOVA technique, The Basic principle of ANOVA, One way ANOVA, Two way ANOVA, Latin-square Design.	8
3	Nature of Intellectual property IPRs- Invention and Creativity - Importance and Protection of Intellectual Property Rights (IPRs) – procedure for grant of patents and patenting under PCT-types of patents-technological research and innovation- international cooperation on IP.	8



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

4	A brief summary of Patents-Copyrights-Trademarks patent rights-licensing and transfer of technology-patent databases-case studies on IPR-Geographical indications-new developments in IPR-protection of IPR rights.	8
5	Interpretation and Report Writing - Meaning of Interpretation, Techniques of Interpretation, Precautions in Interpretation, Significance of Report Writing, Different steps in Writing Report, Layout of the Research Report, Types of Reports, Oral presentation, Mechanics of writing a research Report, Precautions of Writing Research Report.	8

Prescribed Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Research Methodology: Methods and Techniques	C R Kothari, Gaurav Garg	Multicolor(Fourth)	New Age International Publishers	2019
2	An introduction to Research Methodology	Garg, B.L., Karadia, R., Agarwal, F. and Agarwal, U.K	Fourth	RB SA Publishers	2002
3	Handbook of Intellectual property law and practise	Subbaram NR	First	S Viswanathan Printers and Publishing Private Limited	1998

MOOC Course

Sl. No.	Course name	Course Offered By	Year	URL
1.	Understanding Research Methods	Coursera	2024	https://www.coursera.org/learn/research-methods
2.	Fundamentals of Research Methodology	Udemy	2024	https://www.udemy.com/course/fundamentals-of-research-methodology/

Course Outcomes

CO1	Apply appropriate research techniques to conduct studies and present findings.
CO2	Apply ethical principles in conducting research and intellectual property rights, ensuring the integrity and credibility of scientific inquiry.
CO3	Analyze various research problems and their solutions critically.

CO-PO mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3														
CO2	3							2				3			

**B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19**

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

CO3		3												
------------	--	----------	--	--	--	--	--	--	--	--	--	--	--	--

Proposed Assessment Plan (for 50 marks of CIE)

Assessment Tool	No. of Assessments	Marks
Internals	Best of 2	40
QUIZ/AAT	1	10
Total		50

SEE Exam Question paper format

In all five units, two questions are to be asked for 20 marks each as internal choice.

Bloom's Level	Percentage of Questions to be Covered
Remember / Understand	35%
Apply / Analyse	45%
Create / Evaluate	20%

Semester	V		
Course Title:	Big Data Analytics		
Course Code:	23IC5PEBDA	Total Contact Hours:	40 hours
L-T-P:	3-0-0	Total Credits:	3

Unit No.	Topics	Hours
-----------------	---------------	--------------



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

1	Introduction to Big Data and Analytics: Types of Digital Data, Definition of Big data and Analytics, Characteristics of Big Data, Domain Specific Examples of Big Data, Analytics Flow for Big Data, Big Data Stack, Mapping Analytics Flow to Big Data Stack, Case Study: Weather Data Analysis	8
2	CAP theorem, NoSQL: Types of NoSQL databases, MongoDB– Terms used in RDBMS and MongoDB, MongoDB Query Language: CRUD (Create, Read, Update and Delete) operations, Finding documents based on search criteria, Dealing with Null values , Count, Limit, Sort Skip, Aggregate Functions	8
3	Cassandra –Features of Cassandra, Keyspaces, CRUD, Collection, Counters, Time to Live (TTL)operations, Practice Examples. Hadoop: Introducing Hadoop, RDBMS versus Hadoop, Distributed Computing Challenges	8
4	Hadoop Overview, Big Data Storage: HDFS, Batch Analysis: Hadoop and MapReduce, Wordcount Demonstration, Hadoop- MapReduce Examples. Use Case Study- Recommendation Systems: Introduction, A Model for Recommendation Systems, Collaborative Filtering System and Content Based Recommendations.	8
5	Scala: The Basics, Control Structures and functions, Working with arrays, Maps and Tuples Introduction to Data Analysis with Spark: Introduction to Apache Spark, A unified Spark, Users of Spark, Storage layers for Spark. Programming with RDDs: RDD Basics, Creating RDDs, RDD Operations, Passing functions to Spark, Common Transformations and Actions. Spark SQL: Linking with Spark SQL, Using Spark SQL in Applications, Loading and Saving Data, JDBC/ODBC Server, User-defined functions, Spark SQL Performance.	8

Prescribed Text Book

Sl.	Book Title	Authors	Edition	Publisher	Year
-----	------------	---------	---------	-----------	------

**B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19**

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

No.					
1.	Big Data Science & Analytics – A Hands-on Approach	Arshdeep Bahga, Vijay Madiseti	1st	VPT	2019
2.	Big Data and Analytics Seema Acharya,	Seema Acharya, Subjashini Chellappan	1st	Wiley	2010
3.	Scala for the Impatient	Cay S. Horstmann	2nd	Pearson	2016
4.	Learning Spark Lightning-Fast Big Data Analysis	Andy Konwinski, Holden Karau, Matei Zaharia, Patrick Wendell	1st	O'Reilly	2015

Reference Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	A Textbook on Big Data Analytics	Dr. V. Harsha Shastri Mrs. V Sreeprada Mr. T. Udhayakumar Dr. K. R. Ananth	1st	Notion Press	2023
2.	Big Data Analytics, Introduction to Hadoop, Spark, and Machine Learning	Raj Kamal Preeti Saxena	1st	McGraw Hill	2019
3.	Hadoop Operations	Eric Sammer	1st	O'Reilly	2012

MOOC Course

Sl. No.	Course name	Course Offered By	Year	URL
1.	Hadoop Starter Kit	Udemy	2023	Free Hadoop Tutorial - Hadoop Starter Kit Udemy
2.	NPTEL IIT Patna 2023	NPTEL IIT Patna	2023	https://nptel.ac.in/courses/10610418 ,

Course Outcomes*At the end of the course the student will be able to*

CO1	Apply the concepts of NoSQL, Hadoop, Spark for a given task.
CO2	Analyze data analytic techniques for a given problem.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

CO3	Design and develop solutions for given requirements using data analytics mechanisms.
------------	--

CO-PO mapping

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PSO3
CO1	3												3		
CO2		3													
CO3			3		3				1	1					1

Proposed Assessment Plan (for 50 marks of CIE)

Assessment Tool	No. of Assessments	Marks
Internals	2	40
AAT	1	10
Total		50

AAT Plan:

1. Demonstration and installation of HADOOP single node cluster
2. Students should be able to install Cassandra and Spark.
3. Execution of programs and submission of reports.

Sl. No	Week	Activity
•	1 st and 2 nd	Formation of groups. Note: Student groups of size 2 members only
•	3 rd	AAT topic selection by each group
•	4 th	Presentation: Student team and topic introduction by each group
•	5 th , 6 th Design	Design and workflow
•	7 th	Phase 1 - presentation
•	8 th , 9 th , 10 th	Design and Development of the actual algorithm and testing it for various test cases.
•	11 th	Complete code demonstration
•	12 th	AAT Report Preparation

Rubrics used for evaluation(AAT):



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Criteria	Exemplary	Proficient	Partially Proficient	Points
Implementation of the data analysis techniques using appropriate tools	(5-4) Implementation of the data analysis techniques has been done accurately.	(3-2) Implementation of the data analysis techniques has been done appropriately.	(1) Implementation of the data analysis techniques has been done.	___ /4
Testing for various cases	(1) The implemented algorithm works for any given valid input.	(0.75) The implemented algorithm works for almost all valid inputs.	(0.5) The implemented algorithm works for any valid inputs.	___ /1
Application/Relevance	(1) The designed technique has several applications and is relevant in the area of BDA.	(0.75) The designed technique has few applications and is relevant in the area of BDA.	(0.5) The designed technique has few applications and is not very relevant in the area of BDA.	___ /1
Report	(1) Clear and Effective writing and adherence to appropriate style guidelines	(0.75) Writing that is clear and effective for the most part and minor errors in adherence to appropriate style guidelines	(0.5) Unclear and ineffective writing and multiple errors in adherence to appropriate style guidelines	___ /1
Oral communication (presentation)	(1) Clear and effective communication	(0.75) Communication is clear	(0.5) Unclear communication	___ /1



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Participation in Discussions	(1)	(0.75)	(0.5)	___/1
	Provided many good ideas; inspired others; clearly communicated ideas, needs, and feelings.	Participated in discussions; on some occasions, made suggestions.	Listened mainly; Rarely spoke up, and ideas were off the mark.	
Total				___/ 10

SEE Exam Question paper format

In all five units, two questions are to be asked for 20 marks each as internal choice.

Semester	V		
Course Title:	Distributed Systems		
Course Code:	23IC5PEDSY	Total Contact Hours:	40 hours
L-T-P:	3-0-0	Total Credits:	3

Unit No.	Topics	Hours
1	Introduction: Definition–Relation to computer system components –Motivation – Relation to parallel systems – Message-passing systems versus shared memory systems –Primitives for distributed communication Synchronous versus asynchronous executions A model of distributed computations: A distributed program –A model of distributed executions –Models of communication networks –Global state – Cuts –Past and future cones of an event –Models of process communications. Logical Time: A framework for a system of logical clocks –Scalar time –Vector time – Physical clock synchronization: NTP	8
2	Message Ordering & Snapshots Message ordering and group communication: Message ordering paradigms – Asynchronous execution with synchronous communication Synchronous program order on an asynchronous system –Group communication – Causal order (CO) - Total order. Global state and snapshot recording algorithms: Introduction –System model and definitions –Snapshot algorithms for FIFO channels	8

B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

3	Distributed Mutex & Deadlock Distributed mutual exclusion algorithms: Introduction – Preliminaries – Lamport’s algorithm – Ricart - Agrawala algorithm – Suzuki – Kasami’s broadcast algorithm. Deadlock detection in distributed systems: Introduction – System model – Preliminaries – Models of deadlocks – Knapp’s classification of Distributed Deadlock detection algorithm– Mitchell and meritts Algorithms for the single resource model.	8
4	Recovery & consensus Check pointing and rollback recovery: Introduction – Background and definitions – Issues in failure recovery – Checkpoint-based recovery – Log-based rollback recovery – Koo-toueg Coordinated checkpointing algorithm – Juang-Venkateshan Algorithm for asynchronous checkpointing and recovery. Consensus and agreement algorithms: Problem definition – Overview of results – Agreement in a failure – free system – Agreement in synchronous systems with failures.	8
5	P2P & distributed shared memory Peer-to-peer computing and overlay graphs: Introduction – Data indexing and overlays – Chord distributed hash table – Content addressable networks – Tapestry. Distributed shared memory: Abstraction and advantages – Memory consistency models –Shared memory Mutual Exclusion.	8

Prescribed Text Books

Book Title	Authors	Edition	Publisher	Year
Distributed computing: principles, algorithms, and systems	Kshemkalyani, Ajay D., and Mukesh Singhal	1st	Cambridge University Press	2011

Reference Text Books

Book Title	Authors	Edition	Publisher	Year
Distributed Systems Concepts and Design	George Coulouris, Jean Dollimore and Tim Kindberg,	5th	Pearson Education	2012
Distributed Systems: Principles and Paradigms	Tanenbaum A.S., Van Steen M.	2nd	Pearson Education	2007

Course Outcomes

CO1	Apply knowledge of Mutual Exclusion, Deadlock detection algorithms in distributed systems
CO2	Analyze the issues of distributed systems, synchronization issues and global state for distributed systems.
CO3	Apply agreement protocols and fault tolerance mechanisms in distributed systems for peer-to-peer and distributed shared memory systems.

CO PO Mapping

[illegible]

**B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19**

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

CO3	3											2		
-----	---	--	--	--	--	--	--	--	--	--	--	---	--	--

Proposed Assessment Plan (for 50 marks of CIE)

Tool	Remarks	Marks
Internals	Best 2 of 3	40
QUIZ	2	10
Total		50

MOOC Course

Sl. No.	Course name	Course Offered By	Year	URL
1.	Distributed Systems, IIT Patna	NPTEL	2024	https://nptel.ac.in/courses/106106168
2.	DelftX: Modern Distributed Systems	edX	2024	https://www.edx.org/learn/computer-science/delft-university-of-technology-modern-distributed-systems

SEE Exam Question paper format

In all five units, two questions are to be asked for 20 marks each as internal choice.

Semester	V		
Course Title:	Advanced Computer Network		
Course Code:	23IC5PEACN	Total Contact Hours:	40 hours
L-T-P:	3-0-0	Total Credits:	3

Unit No.	Topics	Hours
1	Networking: Datagrams, virtual circuit switching, source routing, bridges and LAN switches, Internetwork, service model, global addresses, datagram forwarding in IP, subnetting and classless addressing, Address Translation (ARP), Host Configuration (DHCP), Error Reporting (ICMP), virtual networks and Tunnels, Distance Vector (RIP), Link State (OSPF), routing areas, inter-domain routing (BGP), IPv6.	8



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

2	End-to-End Protocols: Simple Demultiplexer (UDP), Reliable Byte Stream(TCP), End-to-End Issues, Segment Format, Connecting Establishment and Termination, Sliding Window Revisited, Triggering Transmission-Silly Window Syndrome, Nagle's Algorithm, Adaptive Retransmission-Karn/Partridge Algorithm, Jacobson Karels Algorithm, Record Boundaries, TCP Extensions.	8
3	Congestion Control and Resource Allocation: Issues in resource allocation – network model, taxonomy, evaluation criteria; Queuing discipline – FIFO, Fair Queuing; TCP congestion control – additive increase/multiplicative decrease, slow start, fast retransmit and fast Recovery, Congestion avoidance mechanisms– DECbit, Random Early Detection (RED), Source-based congestion control.	8
4	Network Management: The Infrastructure for Network Management, The Internet-Standard Management Framework, Structure of Management Information: SMI, Management Information Base: MIB, SNMP Protocol Operations and Transport Mappings, Security and Administration, ASN.1, Conclusion.	8
5	Cellular Networks and LTE Technology: The Cellular Concept – System Design Fundamentals: Frequency Reuse, Channel Assignment Strategies, Handoff Strategies, Long-Term Evolution (LTE) Introduction: Architectural Review of UMTS and GSM, The Need for LTE, From UMTS to LTE, From LTE to LTE-Advanced, System Architecture Evolution: High-Level Architecture of LTE, User Equipment	8

Prescribed Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1	Computer Networks: A System Approach	Larry Peterson and Bruce S Davis	5th Edition	Elsevier	2014
2	Data Communications and Networking	Behrouz A. Forouzan,	5th Edition	Tata McGraw Hill	2013
3	An Introduction to LTE	Christopher Cox	2nd Edition	Wiley	2014

Reference Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	High Speed Networks	William Stallings	2nd	Pearson	2002



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

	and Internet: Performance and Quality of Service			Education	
2.	Network Management Principles and Practice	Mani Subramanian	2nd	Pearson,	2010

E-Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year	URL
1.	An Introduction to Computer Networks	Peter L Dordal	Release 2.0.11	Loyola University	Jul 20, 2023	http://intronetworks.cs.luc.edu/current2/ComputerNetworks.pdf

MOOC Course

Sl. No.	Course name	Course Offered By	Year	URL
1.	Advanced Computer Networks	NPTEL/ SWAYAM	2023	https://onlinecourses.nptel.ac.in/noc23_cs35/preview
2.	Introduction to TCP/IP	Coursera	2024	https://www.coursera.org/learn/tcpip

Course Outcomes

At the end of the course, the student will be able to

CO1	Apply fundamental concepts in network and IP addressing in real-world scenario
CO2	Apply computer network concepts to solve protocol layer design issues.
CO3	Analyze the different IP addressing mechanisms, routing, congestion control, and management layers

CO-PO mapping

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3												1		
CO2	3												1		
CO3		3													

Proposed Assessment Plan (for 50 marks of CIE)



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Assessment Tool	No. of Assessments	Marks
Internals	2	40
QUIZ/AAT	2	10
Total		50

SEE Exam Question paper format

In all five units, two questions are to be asked for 20 marks each as internal choice.

Semester	V		
Course Title:	Mini Project in Security		
Course Code:	23IC5PWSEC		
L-T-P:	0-0-2	Total Credits:	2

Guidelines

- **Implementation of Security Techniques:** Under this mini project, students should refer to **recent journal papers (IEEE, Springer, Elsevier, etc.)** focused on security domains such as cryptography, network security, cybersecurity, or secure software development. They are expected to implement the algorithms and technologies mentioned in the selected papers.
- **Group Formation:** Students can form a group with a minimum of two and a maximum of four members.
- **Instructor Guidance:** The instructor assigned to the mini project will guide the students on selecting a relevant domain and technical paper, and will assist them during lab hours in accordance with the timetable. The instructor will also oversee the project work and evaluate the students' progress.

Course Outcomes (CO):

CO1	Analyse security algorithms and techniques as presented in the selected technical paper.
CO2	Design and implement a mini project applying security practices to solve societal and engineering problems in the domain of cybersecurity.
CO3	Participate in team project, present, and create report.

CO-PO-PSO mapping

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1		2													
CO2			3		3								2		3
CO3								1	2	1					



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Rubrics for Mini Project Evaluation

Criteria	Exemplary	Proficient	Partially Proficient	Points
Analysis of Security Techniques	(10-7) All techniques are analysed thoroughly with respect to security requirements.	(6-5) Most techniques are analysed adequately with respect to security requirements.	(4-1) Some techniques are not analysed adequately with respect to security requirements.	_/10
Design of Security Modules	(10-7) All modules are designed appropriately with respect to security needs.	(6-5) Most modules are designed adequately with respect to security needs.	(4-1) Some modules are not designed adequately with respect to security needs.	_/10
Implementation	(10-7) Modules are implemented using appropriate features for all objectives.	(6-5) Modules are implemented for most objectives with appropriate features.	(4-1) Some modules are not implemented in accordance with the design.	_/10
Demonstration of Security Project	(10-7) Demonstrates project functionality with appropriate security features.	(6-5) Demonstrates functionality with most security features.	(4-1) Demonstrates functionality with few security features.	_/10
Report	(10-7) Clear and effective writing with adherence to style guidelines.	(6-5) Clear writing with minor errors.	(4-1) Unclear writing with multiple errors.	_/10
Total				_/50

Weekly Activities and Delivery

Week	Activity	Content Deliverables by Assigned Teacher	Lab experiments to implement by students
1st	Group formation (2-4 students)	Introduction to security domains and guidance on selecting technical papers.	- Write a program that encrypts and decrypts a message using the Caesar cipher technique. Allow the user to input the shift value for encryption and decryption. - Develop a program to encrypt and decrypt text using the Vigenère cipher. The program should take a keyword and a plaintext message as inputs.
2nd	Topic selection by each group	Discuss the security techniques/algorithms from the selected paper.	Create a program to perform encryption and decryption using the AES algorithm. The program should allow the user to choose between 128-bit, 192-bit, or 256-



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

			bit keys.
3rd	Presentation: Topic introduction by each group	Verify the study of existing and proposed security techniques from the paper.	4. Implement a basic RSA algorithm to encrypt and decrypt messages. The program should include key generation, encryption, and decryption functions.
4th - 6th	25% of project work completion	Review the progress and provide feedback.	5. Write a program to generate a SHA-256 hash of a given message. This will demonstrate how to create a fixed-size hash from variable-length input data.
7th	Presentation: 25% working model	Evaluate and review the progress made so far.	6. Develop a program that implements HMAC using a cryptographic hash function like SHA-256. The program should take a message and a secret key as inputs and produce the HMAC.
8th - 9th	50% of project work completion	Continued review and guidance on project development.	7. Implement the Diffie-Hellman key exchange algorithm to demonstrate how two parties can securely share a secret key over a public channel.
10th	Discussion on results	Validate the results obtained and provide further guidance.	8. Create a program that uses RSA to sign a message digitally. The program should include functions for signing the message with a private key and verifying the signature with the corresponding public key.
11th	Complete project work	Final presentation and evaluation.	Complete project work
12th	Project report preparation	Provide feedback on report writing and final documentation.	Project report preparation

SEE Exam (50 Marks)

Evaluation of Projects will be carried out by an External examiner along with internal faculty.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Semester	VI		
Course Title:	Ethical Hacking		
Course Code:	23IC6PCEHG	Total Contact Hours:	40 hours
L-T-P:	3-0-1	Total Credits:	4

Unit No.	Topics	Hours
1	Introduction to Hacking: Important Terminologies, Categories of Penetration Testing: Black Box, White Box, Grey Box, Types of Penetration test, Writing Reports, Structure of a Penetration Testing Report, Vulnerability Assessment Summary, Risk Assessment, Methodology. Footprinting and Social Engineering: Using Web Tools for Footprinting, Conducting Competitive Intelligence, Using Domain Name System Zone Transfers, Introduction to Social Engineering- The Art of Shoulder Surfing, Dumpster Diving, Piggybacking, Phishing	8
2	Port Scanning: Introduction to port scanning, Using Port-Scanning Tools- Nmap, Unicornscan, Nessus and OpenVAS, Conducting Ping Sweeps- Fping, Hping, Crafting IP Packets Enumeration: Introduction to Enumeration, Enumerating Windows Operating Systems- NetBIOS Basics, NetBIOS Enumeration Tools, Additional Enumeration Tools, Enumerating the NetWare Operating System, Enumerating the *nix Operating system.	8
3	Hacking the Network Network Infrastructure: Network Infrastructure Vulnerabilities, Choosing Tools, Scanning, Poking, and Prodding, Common Router, Switch, and Firewall Weaknesses, General Network Defenses. Wireless LANs: Understanding the Implications of Wireless, Choosing Your Tools, Wireless LAN Discovery, Wireless Network Attacks and Countermeasures- Rogue wireless devices, Countermeasures against rogue wireless devices, MAC spoofing, Countermeasures against MAC spoofing, Countermeasures against DoS attacks.	8
4	Hacking Operating Systems Windows: Windows Vulnerabilities, Choosing Tools, Information Gathering, Null Sessions, Share Permissions, Missing Patch Exploitation, Authenticated Scans. Linux: Linux Vulnerabilities, Choosing Tools, Information Gathering, Unneeded and Unsecured Services, .rhosts and hosts.equiv Files, NFS, File Permissions, Buffer Overflows, Physical Security, General Security Tests, Patching Linux	8
5	Hacking Applications Communication and Messaging Systems: Messaging System Vulnerabilities, E- Mail Attacks, Instant Messaging, Voice over IP Web Sites and Applications: Choosing Your Web Application Tools, Web	8



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

	Vulnerabilities, Best Practices for Minimizing Web Security Risks Databases and Storage Systems: Databases, Best Practices for Minimizing Database Security Risks, Storage Systems, Best Practices for Minimizing Storage Security Risks	
--	--	--

Prescribed Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Ethical Hacking and Penetration Testing Guide	Rafay Baloch	First	Tata Mc Graw Hill Publishers	2014
2.	Hands-On Ethical Hacking and Network Defense	Michael T. Simpson, Kent Backman, and James E. Corley	2nd Edition	Delmar Cengage Learning	2010
3.	Hacking For Dummies	Kevin Beaver	3rd Edition	Wiley Publishing, Inc.	2010

Reference Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1	Hacking Exposed Network Security Secrets & Solutions	Stuart McClure, Joel Scambray and Goerge Kurtz		Tata Mc Graw Hill Publishers	2010

E-Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year	URL
1.	Learn Ethical hacking from scratch	Zaid Sabih	-	Packt Publisher	-	https://www.packtpub.com/free-ebook/learn-ethical-hacking-from-scratch/9781788622059?srsid=AfmBOortSRnOYATgQWVla7E_FrwuNj7tFr708g72gacuHb44YFZD_gbW

MOOC Course

Sl. No.	Course name	Course Offered By	Year	URL
1.	Ethical Hacking	NPTEL	2022	https://onlinecourses.nptel.ac.in/noc22_cs13/preview



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Course Outcomes

At the end of the course, the student will be able to

CO1	Apply the concepts of ethical hacking and exposure to tools and methods.
CO2	Analyze the aspects of security, data gathering, foot-printing and system hacking.
CO3	Design plans to perform penetration testing for real-world scenarios.
CO4	Conduct experiments to implement cyber attacks and its countermeasures.

CO-PO mapping

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PSO 3
CO1	3				2								2		3
CO2		3			2			3							3
CO3			3					3					2		3
CO4				3	3								2		3

Proposed Assessment Plan (for 50 marks of CIE)

Assessment Tool	No. of Assessments	Marks
Internals	2	20
QUIZ/AAT	1	5
Lab	CIE + 1 Test	25
Total		50

SEE Exam Question paper format

In all five units, two questions are to be asked for 20 marks each as internal choice.

Laboratory Plan:

Experiment	Name of the Experiment
1	Perform passive footprinting on a target network using tools like whois, Nslookup, and Google Hacking Database. Gather information about the network and its assets.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

2	Create a script to automate the scanning of a network using Nmap, identify open ports, and classify potential vulnerabilities based on the scan results.
3	Simulate an SQL Injection attack on a vulnerable web application using SQLMap/BurpSuite, then implement defenses to prevent such attacks.
4	Implement a dictionary-based password cracking attack using John the Ripper and analyze the effectiveness of different password policies.
5	Perform a session hijacking attack using Wireshark to capture session tokens, then implement secure session management to prevent hijacking
6	Simulate a phishing attack using the Social Engineering Toolkit (SET) and discuss how to detect and prevent social engineering attacks.
7	Perform penetration testing on a wireless network using Aircrack-ng/Wireshark, identify vulnerabilities, and implement WPA3 encryption for enhanced security.
8	Exploit common vulnerabilities in a web application using Burp Suite, including XSS and CSRF attacks, and implement secure coding practices to mitigate these issues.
9	Use Metasploit to exploit known vulnerabilities in a virtualized environment, including post-exploitation tasks such as privilege escalation and data extraction.
10	Create a simple ransomware in Python that encrypts files on a system, then develop a decryption tool and discuss the ethical and legal implications.
11	Simulate a MITM attack using Ettercap, intercept and modify communications, and implement HTTPS to secure the communication channel.
12	Organize a red team (attack) vs. blue team (defense) exercise, where the red team attempts to breach the system while the blue team defends it.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

--	--



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Semester	VI		
Course Title:	Cloud Computing and Security		
Course Code:	23IC6PCCCS	Total Contact Hours:	40 hours
L-T-P:	2-1-0	Total Credits:	3

Unit No.	Topics	Hours
1	Introduction to Cloud Computing: Introduction, Characteristics of cloud computing, Cloud Models: Service Models, Deployment Models, Cloud Services Examples: IaaS, PaaS, SaaS, Cloud based services and applications, Cloud Concepts and Technologies.	8
2	Virtualization for Cloud: Need for Virtualization, Pros and cons of Virtualization, Types of Virtualization, System VM, Virtual Machine monitor, Virtual Machine properties, Interpretation and binary translation, HLL VM, Hypervisors, Xen, KVM, VMWare, Live VM migration steps and performance effects	8
3	Cloud Platform Architecture over Virtualized Data Centers: Cloud Computing service Models: Public, private and Hybrid clouds, Data Center Design and Interconnection networks, Architectural Design of Compute and storage clouds, Public Cloud Platforms: GAE, AWS and Azure, Inter-cloud Resource Management, Cloud Security and Trust Management.	8
4	Inter-Cloud Resource Management: Extended Cloud Computing services, Resource provisioning and platform deployment, Virtual Machine creation and Management, Global Exchange of Cloud resources. Cloud Security and Trust Management: Cloud security defense strategies, Distributed intrusion/ anomaly detection, data and software protection techniques, Reputation guided protection of data centers.	8
5	Cloud Computing Software Security Fundamentals: Relevant Cloud Security Design Principles, Secure Development Practices, Approaches to Cloud Software Requirements Engineering, Cloud Security Policy Implementation and Decomposition, NIST 33 Security Principles Secure Cloud Software Testing: Testing for Security Quality Assurance, Cloud Penetration Testing, Regression Testing, General Principle and Practices, Outsourcing, CSA STAR.	8

**B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19**

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)**Prescribed Text Book**

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Cloud Computing: A Hands- on Approach	Arshdeep Bagha, Vijay Madiseti	1st Edition	University Press	2014
2.	Cloud Security- A Comprehensive Guide to secure cloud computing	Ronald L. Krutz and Russell Dean Vines	1st Edition	Wiley	2010
3.	Distributed and cloud Computing from Parallel Processing to the Internet of Things	Kai Hwang, Geoffrey C. Fox, Jack J Dongarra	1st Edition	Morgan Kaufmann, Elsevier	2013
4.	Virtual Machines	James E Smith, Ravi Nair	1st Edition	Morgan Kaufmann Publishers	2006

Reference Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1	Cloud Computing – Principles and Paradigms	Rajkumar Buyya, Broberg Andrzej Goscinski	1st Edition	WILEY: A JOHN WILEY & SONS, INC.,	2011

E-Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year	URL
1.	Cloud Computing	Dr. Kumar Saurabh	2 nd Edition	WILEY INDIA.	2012	https://www.amazon.in/Cloud-Computing-2edKumar-Saurabhebook/dp/B01M02ECEH

MOOC Course

Sl. No.	Course name	Course Offered By	Year	URL
1.	Cloud Application Security	Coursera	2021	https://www.mooc-list.com/course/cloud-application-security-coursera

Course Outcomes*At the end of the course, the student will be able to*

CO1	Apply fundamental concepts of Cloud Computing
CO2	Analyze technological concepts of Cloud Computing.
CO3	Apply security standards, policies and compliance requirements to cloud applications.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

CO4	Implement a resource scheduling algorithm for a given scenario using a suitable simulator.
------------	--

CO-PO mapping

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	3												2		
CO2		3													
CO3	3												2		
CO4			3		3				1	1					3

Proposed Assessment Plan (for 50 marks of CIE)

Assessment Tool	No. of Assessments	Marks
Internals	2	40
AAT	1	10
Total		50



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Tutorial Plan:

Tutorial	Unit	Topic
1	I	Demonstration of SaaS application(Demonstrate CRM application suitable platform)
2	III	Demonstration of PaaS application (Create a Trailblazer app on sales force trailhead platform)
3	II	a. Create a virtual machine using VMWare Workstation PRO12 (VMware PlayStation) b. Install Python on created a virtual machine and run a simple program.
4	II	Installation of CloudSim Simulator
5	II	Demonstration and program implementation on CloudSim Simulator
6	V	Demonstration of Dockers and Kubernetes(Industry Connect)
7	III	Complete Introduction to Cloud Course on AWS Educate Learner Dashboard Page (awseducate.com)
8	IV	Complete Getting Started with Security on AWS Educate Learner Dashboard Page (awseducate.com)
9	II	Complete Getting Started with Compute on AWS Educate Getting Started with Compute (instructure.com)
10	V	Complete Getting Started with Serverless on AWS Educate Learner Dashboard Page (awseducate.com)
11	III, IV, V	Create an application for data analysis (Use any front end tool and NoSQL Database as backend), Dockize the same and host it on a chosen cloud (AWS, Google cloud etc)
12	III, IV, V	Create an application for data analysis (Use any front end tool and NoSQL Database as backend), Dockize the same and host it on a chosen cloud (AWS, Google cloud etc)

Alternate Assessment Tool Plan:
Students should form a group of 3 or 4 to analyse a suitable Simulator and implement a scheduling algorithm among VMs as a mini project. Some of the assignments allotted to students are under the topics: Students have to make an analysis of the output and generate graphs for the following:
1. Resource Utilization Graph
2. Response time of each

task

3. Waiting time of each task

The plan of activities is as follows:

Sl No.	Week	Activity
1	1 st and 2 nd	Formation of groups. Note: Student groups of size 2 or 3 or 4 to be formed within the class.
2	3 rd	Selection of simulator by each group
3	4 th	Selection of a scheduling algorithm
4	5 th	Design of the scenario



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

5	6 th 7 th ,8 th ,9 th	Implementation and testing
6	10 th ,11 th	Report preparation
7	12 th ,13 th	Presentation of the project (Demonstration, report and PPT)

Evaluation Rubrics: 10 Marks

Criteria	Exemplary	Proficient	Partially Proficient	Points
Implementation	(4) Ideal implementation for all the set objectives	(2) Appropriate implementation for most of the set objectives	(1) Concern to implementation for a few set objectives	___ / 4
Report	(2) Clear and Effective writing and adherence to appropriate style guidelines	(1) Writing that is clear and effective for the most part and minor errors in adherence to appropriate style guidelines	(0.5) Unclear and ineffective writing and multiple errors in adherence to appropriate style guidelines	___/2
Oral communication (presentation)	(2) Clear and effective communication	(1) Communication is clear	(0.5) Unclear communication	___/2
Team Work	(1) Actively participated as a team member	(0.5) Partially participated as a team member	(0) No participation	___/1
Participation in the Discussions	(1) Provided many good ideas; inspired others; clearly communicated ideas, needs, and feelings.	(0.75) Participated in discussions; on some occasions, made suggestions.	(0.5) Listened mainly; Rarely spoke up, and ideas were off the mark.	___/1
Total				___/10

Tutorial Plan:

Tutorial No.	Unit No.	Topic
1	I	Demonstration of SaaS application (Demonstrate CRM application suitable platform)



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

2	III	Demonstration of PaaS application (Create a Trailblazer app on sales force trailhead platform)
3	II	a. Create a virtual machine using VMWare Workstation PRO12 (VMware PlayStation) b. Install Python on created a virtual machine and run a simple program.
4	II	Installation of CloudSim Simulator
5	II	Demonstration and program implementation on CloudSim Simulator
6	V	Demonstration of Dockers and Kubernetes (Industry Connect)
7	III	Complete Introduction to Cloud Course on AWS Educate Learner Dashboard Page (awseducate.com)
8	IV	Complete Getting Started with Security on AWS Educate Learner Dashboard Page (awseducate.com)
9	II	Complete Getting Started with Compute on AWS Educate Getting Started with Compute (instructure.com)
10	V	Complete Getting Started with Serverless on AWS Educate Learner Dashboard Page (awseducate.com)
11	III, IV, V	Create an application for data analysis (Use any front-end tool and NoSQL Database as backend), Dockize the same and host it on a chosen cloud (AWS, Google cloud etc.,)
12	III, IV, V	Create an application for data analysis (Use any front-end tool and NoSQL Database as backend), Dockize the same and host it on a chosen cloud (AWS, Google cloud etc.,)

SEE Exam Question paper format

In all five units, two questions are to be asked for 20 marks each as internal choice.

Sem	VI		
Course Title:	Blockchain Technology		
Course Code:	23IC6PCBCT	Total Contact Hours:	40
L-T-P:	3-0-1	Total Credits:	4

Unit No.	Topics	Hrs
1	Blockchain Essentials: The history of Blockchain and Bitcoin, The Growth of Blockchain Technology, Blockchain- Definition, Architecture, Generic elements of a Blockchain, Benefits, Features and Limitations, Types of Blockchain, Consensus. Decentralization: Decentralization using Blockchain, Decentralized Applications Cryptographic Constructs in Blockchain: Cryptographic primitives- Hash function, ECC Digital Signature, Zero Knowledge proof, Different types of digital signatures. Applications of Cryptographic Hash Functions: Merkel Trees, Patricia Trees, Distributed Hash Table.	8



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

2	Consensus Algorithms: Introducing the Consensus Problem, Analysis and Design, Classification, Algorithms, Choosing an Algorithm. Transactions: Transaction Life Cycle, Genesis block, Mining Bitcoin: Overview, Cryptographic Keys, Bitcoin Network, Wallets, Bitcoin Payments, Innovation in Bitcoin, Alternative coins- Introduction, Theoretical Foundations- Alternative Proof of Work.	8
3	Ethereum : An Overview, Ethereum Network, Components of Ethereum Ecosystem, Ethereum Virtual Machines Smart Contracts: History, Definition, Deploying Smart Contracts. Solidity: Variables, Data types, Control Structures, Events, Inheritance, Libraries, Functions, Error Handling. Smart Contract Security: Reentrancy, Arithmetic Over/Underflows	8
4	Decentralized Applications (DApps): What Is a DApp, A Basic DApp Example, Further Decentralizing the Auction DApp, Storing the Auction DApp on Swarm, The Ethereum Name Service (ENS), From App to DApp.	8
5	Hyperledger: Architecture, Projects under Hyperledger, Hyperledger Fabric, Hyperledger Sawtooth. Tokenization: Tokenization on a blockchain, Types of tokens, Process of Tokenization, Token offerings	8

Prescribed Text Book:

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1	Mastering Blockchain	Imran Bashir	3rd	Packt	2020
2	Mastering Ethereum Building Smart Contracts and DApps	Andreas M. Antonopoulos and Dr. Gavin Wood	First Edition	O'Reilly	2018

MOOC Course:

Sl. No.	Course Name	Course Offered by	URL
1	Blockchain Specialization Course 1: Blockchain Basics Course 2: Smart contracts Course 3: Decentralized Applications Course 4: Blockchain Platforms	Coursera	https://www.coursera.org/specializations/blockchain
2.	Blockchain architecture, Design and Use cases	NPTEL	https://nptel.ac.in/courses/106/105/106105184/

Course Outcomes (Co):

CO1	Apply the principles of Blockchain and Cryptocurrency for a given application.
CO2	Analyze the various protocols and mining techniques in Blockchain.
CO3	Design Blockchain based solution for a given scenario.
CO4	Conduct practical experiments to solve problems using Blockchain technology.

CO-PO-PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3												1		
CO2		1													
CO3			3		2									2	3



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

CO4				2	3				1	1						3
-----	--	--	--	---	---	--	--	--	---	---	--	--	--	--	--	---

Proposed Assessment Plan (for 50 marks of CIE)

Tool	Remarks	Marks
Internals	2	20
Quiz	1	05
Lab	CIE + 1 Test	25
Total		50

SEE Exam Question paper format

In all five units, two questions are to be asked for 20 marks each as internal choice.

Laboratory Plan

Instructions to Students to be followed in each lab:

1. Each Student should write down the program in the observation book and get it evaluated by the respective lab faculty in-charge and then execute the program.
2. Each Student should bring the lab record with the programs and output written for the programs completed in their respective previous week and get it evaluated by the lab faculty in-charge. In the record book students should - Handwrite the Program - and Output.
3. Students have to practice following list of programs and additional programming exercises will also be given in the lab.

Program No	Program Details
1	Write a basic Solidity smart contract to understand the fundamentals of the language.
2	Implement a Solidity contract that uses arrays and control structures (if-else, loops).
3	Write a Solidity contract using different types of functions (view, pure, fallback) and function modifiers.
4	Implement a Solidity contract using mappings and structs to create a simple record-keeping system.
5	Install MetaMask, connect to an Ethereum testnet, and interact with a deployed smart contract.
6	Set up Truffle and Ganache, compile and deploy a Solidity contract on a local blockchain.
7	Implement a Merkle tree and basic blockchain focusing on state alteration and block creation.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

8	Create and deploy an ERC20 token on Ethereum. Implement a blockchain with a mining process in Python, simulating the Proof-of-Work consensus algorithm
9	Implement a basic peer-to-peer blockchain network in Python.
10	Implement a blockchain with a mining process in Python, simulating the Proof-of-Work consensus algorithm
11	Install Hyperledger Fabric and Composer, then deploy and test a sample blockchain application.

Bloom's Level	Percentage of Questions to be Covered
Remember / Understand	25%
Apply / Analyze	50%
Create / Evaluate	25%



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Semester	VI		
Course Title:	Software Engineering & Agile Methodologies		
Course Code:	23DC6PCSEA	Total Contact Hours:	30 hours
L-T-P:	2-0-0	Total Credits:	2

Unit No.	Topics	Hours
1	Introduction - Overview of Software Development Projects, Emergence of Software Engineering, Software Metrics Software Life Cycle Models - Basic Concepts, Waterfall Model and its Extensions, Rapid Application Development, Agile Development Models: Essential Ideas Behind Agile Models, Agile vs. Other Models, Spiral Model, Comparison of Different Life Cycle Models	6
2	Requirements Analysis and Specification - Requirements Gathering and Analysis, Software Requirements Specification (SRS) Case Study - IEEE SRS Software Design - Overview of the Design Process, Characteristics of Good Software Design: Cohesion and Coupling, Approaches to Software Design Function-Oriented Software Design - Overview of SA/SD Methodology, Structured Analysis: Developing the DFD Model of a System, Structured Design and Detailed Design	6
3	Software Project Management - Project Planning, Metrics for Project Size Estimation, Project Estimation Techniques: Empirical Estimation Techniques, COCOMO (A Heuristic Estimation Technique), Scheduling and Team Structures	6
4	Understanding Agile - What is Agile?, The Agile Manifesto and Principles, Why Agile Works Better than Traditional Models Kanban and Lean - Introduction to Kanban Method, Lean Principles in Agile Jira Fundamentals - Overview of Jira: Project Boards, Enrich Issues, Kanban Boards, Scrum Projects, Quick Search and Basic Search, JQL (Jira Query Language), Filters, Epics, Dashboards	6
5	Understanding XP - The XP Life cycle, The XP Team, XP Concepts, Software Configuration Management - Configuration Management Process, Version Control Systems, Change Management and Control	6



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Prescribed Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Fundamentals of Software Engineering	Rajib Mall	5th	PHI Learning	2018
2.	Agile Foundations: Principles, Practices, and Frameworks	Peter Measey	1st	BCS Learning & Development Limited	2015
3	Atlassian Jira Service Desk A Complete Guide	Gerardus Blokdyk	1st	5STARCooks	2020
4	The Art of Agile Development	James Shore & Shane Warden	2nd	O'Reily	2007

Reference Book

Sl. No	Book Title	Authors	Edition	Publisher	Year
1.	Essential Scrum: A Practical Guide to the Most Popular Agile Process,	Kenneth Rubin	-	Pearson	2017
2.	Hands-On Agile Software Development with JIRA: Design and manage software projects using the Agile methodology	David Harned	-	Packt	2018

MOOC Course

Sl. No.	Course name	Course Offered By	Year	URL
1.	Agile Software Development Framework	IIT-K	2024	https://ict.iitk.ac.in/courses/agile-software-development-framework-scrum/

Course Outcomes

At the end of the course the student will be able to

CO1	Apply the principles of software engineering frameworks and processes.
CO2	Apply agile practices and software engineering techniques to manage and develop software projects using agile tools.
CO3	Analyze different software development models and agile practices to select the most effective approach for specific project requirements.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

CO-PO mapping

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO 1	3												3		
CO 2	3				1						3		3		
CO 3		3											3		

Proposed Assessment Plan (for 50 marks of CIE)

Assessment Tool	No. of Assessments	Marks
Internals	2	40
QUIZ/AAT	2	10
Total		50

SEE Exam Question paper format

In all five units, two questions are to be asked for 20 marks each as internal choice.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Semester	VI		
Course Title:	Social Network Analysis		
Course Code:	23IC6PESNA	Total Contact Hours:	40 hours
L-T-P:	3-0-0	Total Credits:	3

Unit No.	Topics	Hours
1	Introduction: The Social Networks Perspective, Historical and Theoretical Foundations, Fundamental Concepts in Network Analysis, Distinctive Features. Social Network Data: What are Network Data? Boundary Specification and Sampling, Types of Networks, Network Data, Measurement and Collection. Graph Theory and Social Networks: Graphs: Basic Definitions, Paths and Connectivity, Distance and Breadth-First Search.	8
2	Strong and Weak Ties: Triadic Closure, The Strength of Weak Ties, Tie Strength and Network Structure in Large-Scale Data, Tie Strength, Social Media, and Passive Engagement, Closure, Structural Holes, and Social Capital. Networks in Their Surrounding Contexts: Homophily, Mechanisms Underlying Homophily: Selection and Social Influence, Affiliation.	8
3	Positive and Negative Relationships: Structural Balance, Characterizing the Structure of Balanced Networks. Information Networks and the World Wide Web: The Structure of the web: The World Wide Web, Information Networks, Hypertext, and Associative Memory, The Web as a Directed Graph, The Bow-Tie Structure of the Web.	8
4	Link Analysis and Web Search: Searching the Web: The Problem of Ranking, Link Analysis using Hubs and Authorities, PageRank. Cascading Behavior in Networks: Diffusion in Networks, Modelling Diffusion through a Network, Cascades and Clusters, Diffusion, Thresholds, and the Role of Weak Ties.	8
5	The Small-World Phenomenon: Six Degrees of Separation, Structure and Randomness, Decentralized Search, Modeling the Process of Decentralized Search, Core-Periphery Structures and Difficulties in Decentralized Search.	8



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Prescribed Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Social Network Analysis: Methods and Applications.	Stanley Wasserman, Katherine Faust	1st	Cambridge University Press	2012
	Networks, Crowds, and Markets: Reasoning about a Highly Connected World	David Easley, Jon Kleinberg.	1st	Cambridge University Press	2010

Reference Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Social Network Analysis	John Scott	3rd Edition	SAGE publications Ltd.,	2012
2.	Understanding-Social-Networks-Theories-Concepts-and findings	Charles Kadushin	1st Edition	Oxford university press	2012

E-Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year	URL
1.	Social Network Analysis theory and applications.	Dr. John T.Bell	-	University of Illinois Chicago	2006 & 2013	https://www.cs.uic.edu/~jbell/CourseNotes/OperatingSystems/index.html

MOOC Course

Sl. No.	Course name	Course Offered By	Year	URL
1.	Social Network Analysis	NPTEL	2024	https://onlinecourses.nptel.ac.in/noc24_cs90/preview

Course Outcomes

At the end of the course the student will be able to

CO1	Apply concepts of social networks for a given scenario
CO2	Analyze data connectivity problems by using social network algorithms
CO3	Analyze the behavior of the social network model for a given scenario

CO-PO mapping



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3												2		
CO2		3													
CO3		3													

Proposed Assessment Plan (for 50 marks of CIE)

Assessment Tool	No. of Assessments	Marks
Internals	2	40
QUIZ/AAT	2	10
Total		50

SEE Exam Question paper format

In all five units, two questions are to be asked for 20 marks each as internal choice.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Semester	VI		
Course Title:	Advanced Cryptography		
Course Code:	23IC6PEACR	Total Contact Hours:	40 hours
L-T-P:	3-0-0	Total Credits:	3

Unit No.	Topics	Hours
1	Transport-Level Security: Web Security Considerations, Transport Layer Security, HTTPS, Secure Shell (SSH), Wireless Network Security: Wireless Security, Mobile Device Security, IEEE 802.11 Wireless LAN Overview, IEEE 802.11i Wireless LAN Security	8
2	Electronic Mail Security Internet Mail Architecture, Email Formats, Small Threats and Comprehensive Email Security, S/MIME, Pretty Good Privacy (PGP), DNSSEC, DNS-Based Authentication of Named Entities (DANE), Sender Policy Framework, DomainKeys Identified Mail, Domain-Based Message Authentication, Reporting, and Conformance, IP Security IP Security Overview, IP Security Policy, Encapsulating Security Payload, Combining Security Associations, Internet Key Exchange, Cryptographic Suites, Key Terms, Review Questions, and Problems	8
3	Zero-Knowledge Protocols Introduction to Zero-Knowledge Protocols, The main scenario of a ZKP - the digital cave, Non-interactive ZKPs, Schnorr's interactive ZKP, An introduction to zk-SNARKs - spooky moon math, ZK-SNARKs in Zcash cryptocurrency, One-round ZKP, ZK13 - A ZKP for authentication and key exchange,	8
4	Homomorphic Encryption and the Crypto Search Engine, Crypto Search Engine, Introduction to CSE – homomorphism, Partial homomorphism in RSA, Analysis of homomorphic encryption and its implications, Math and logic behind search engines, Introduction to trees – graph theory, Huffman code, Index, Hash and Boolean logic, CSE explained, The innovation in CSE, Computational analysis on CSE, Example of computational brute-force cracking, Applications of CSE.	8
5	Q-Mechanics and Q-Cryptography, Introduction to Q-Mechanics and Q-Cryptography, An imaginary experiment to understand the elements of Q-Mechanics, Q-Cryptography, Quantum key distribution - BB84, Quantum computing, Shor's algorithm, Quantum Fourier Transform, Post-Q-Cryptography,	8

Prescribed Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
---------	------------	---------	---------	-----------	------



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

1.	Cryptography and Network Security Principles and practice	William Stallings	7th	Pearson Education Asia	2017
2.	Cryptography Algorithms	Massimo Bertaccini	1 st	Packt Publishing	2022

Reference Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Cryptography and Network Security	Behrouz A. Forouzan and Debdeep Mukhopadhyay	2nd	Tata McGraw Hill	2013
2	Introduction to Modern Cryptography	Jonathan Katz and Yehuda Lindell	3 rd	Chapman & Hall/CRC	2021
3.	Cryptography: Theory and Practice	Stinson. D.	3rd	Chapman & Hall/CRC	2012
4.	Cryptography and Network Security	Atul Kahate	3rd	Tata McGraw-Hill Publishing	2008

E-Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year	URL
1.	Cryptograp hy and Network Security. Principles and Practice	William Stallings	3rd	Pearson Education	2007	http://williamstallings.com/Cr ypto3e.html
2.	Handbook of Applied Cryptograp hy	Menez, van Oorschot, Vanstone	ISBN: 0-8493-8523-7	CRC Press	2001	http://www.cacr.math.uwaterloo.ca/hac/



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

MOOC Course

Sl. No.	Course name	Course Offered By	Year	URL
1.	Cryptography and Network Security	NPTEL	2022	https://onlinecourses.nptel.ac.in/noc22_cs90/prview

Course Outcomes

At the end of the course the student will be able to

CO1	Apply advanced cryptographic techniques to protect data in various communication networks.
CO2	Analyze different security protocols and cryptographic methods to understand their effectiveness and limitations in ensuring data security.
CO3	Implement and demonstrate the use of advanced cryptographic techniques.

CO-PO mapping

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	3												1		
CO2		3													
CO3			3		3				1	1					3

Proposed Assessment Plan (for 50 marks of CIE)

Assessment Tool	No. of Assessments	Marks
Internals	2	40
AAT	1	10
Total		50

Alternate Assessment Tool Plan:

Students are supposed to develop an advanced cryptographic algorithm (using C/C++ preferably) without using libraries or built-in functions. Code demonstration along with a report has to be submitted.

Sl. No	Week	Activity
1.	1st and 2nd	Formation of groups. Note: Student groups of size 2 members only
2.	3rd	AAT topic selection by each group
3.	4 th	Presentation: Student team and topic introduction by each group
4.	5th, 6th	Design the workflow along with Front-end Design
5.	7 th	Presentation on Front-end Design of the application
6.	8 th, 9th, 10th	Design and Development of the actual algorithm and testing it for various test cases.
7.	11th	Complete code demonstration



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

8.	12th	AAT Report Preparation
----	------	------------------------

Rubrics used for evaluation (AAT):

Criteria	Exemplary	Proficient	Partially Proficient	Points
User Interface / Front End Design OR Tool Usage	(1) The designed application has an exceptional design, attractive and usable interface. It is easy to locate all important elements.	(0.75) The designed application has an attractive design and usable interface. It is easy to locate all important elements.	(0.5) The designed application has a usable design interface, but may appear busy or boring. It is easy to locate most of the important elements.	___ / 1
Implementation of the Algorithm OR Implementation done in the Tool	(4) Implementation of the algorithm has been done accurately without the usage of any library functions.	(2.5) Implementation of the algorithm has been done appropriately without the usage of any library functions.	(1.5) Implementation of the algorithm has been done with usage of a few library functions.	___ / 4
Testing for various cases	(1) The implemented algorithm works for any given valid input.	(0.75) The implemented algorithm works for almost all valid inputs.	(0.5) The implemented algorithm works for any valid inputs.	___ / 1
Application/Relevance	(1) The designed algorithm has several applications and is relevant in the area of cryptography.	(0.75) The designed algorithm has few applications and is relevant in the area of cryptography.	(0.5) The designed algorithm has few applications and is not very relevant in the area of cryptography.	___ / 1

**B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19**

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Report	(1) Clear and Effective writing and adherence to appropriate style guidelines	(0.75) Writing that is clear and effective for the most part and minor errors in adherence to appropriate style guidelines	(0.5) Unclear and ineffective writing and multiple errors in adherence to appropriate style guidelines	___/1
Oral communication (presentation)	(1) Clear and effective communication	(0.75) Communication is clear	(0.5) Unclear communication	___/1
Participation in Discussions	(1) Provided many good ideas; inspired others; clearly communicated ideas, needs, and feelings.	(0.75) Participated in discussions; on some occasions, made suggestions.	(0.5) Listened mainly; Rarely spoke up, and ideas were off the mark.	___/1
Total				___/ 10

SEE Exam Question paper format

In all five units, two questions are to be asked for 20 marks each as internal choice.

Semester:	VI		
Course Title:	Major Project Phase-1		
Course Code:	23IC6PWPP1		
L-T-P:	0-0-2	Total Credits:	2

Guidelines:

1. Students should form teams to carry out the project. The size of the teams can comprise of a minimum of three students and maximum of four students.
2. Teams are free to choose their Internal Guide or will be assigned an Internal Guide by the Department Coordinator.
3. Teams can carry out their project in-house or in a reputed organization (which has to be approved by the Internal Guide). Students taking up industry projects can do so with the condition that they are allowed to demonstrate their project work on the college campus.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

4. Identification of a problem which is feasible and innovative based on the current state of art technology and having relevance and social impact, considering the boundaries of societal, environmental and ethical issues.
5. The project should address any one or more than one of the Seventeen Sustainable Development Goals (SDG) given by United Nations. The project should belong to the category of either "Community Oriented Project" or "Multidisciplinary Project" or both.
6. Survey of literature related to the identified problem to make a feasibility study and identify the project requirements. Prepare and submit a synopsis of your project to your respective Guides.
7. Based on the literature review, preparation of review paper and publishing it.
8. Evolve a high-level design/system level architecture and identify the various implementable modules with their input/output needs.
9. Preparation and submission of Project Phase1 technical report.
 - In-house project teams should meet their Guides weekly and update about the progress of the projects, whereas the industry project students should meet their Guides once in fifteen days.
 - CIE evaluation: Carried out twice in a semester by an internal panel comprising of Internal Guide and at least three other faculty members of the department.
 - SEE evaluation: Conducted at the end of the sixth semester by both internal guide and an external examiner from other institutions. Evaluation is based on the team presentation, project report and publication.

Note: CIE and SEE will be evaluated for 100 Marks and will be reduced to 50 Marks.

Course Outcomes (COs):

CO1	Identify a real-life/engineering problem, utilize prior knowledge, and conduct extensive investigation across diverse sources, in addressing the challenges associated with the problem.
CO2	Competence in applying the software engineering principles in planning, formulating an innovative design/ approach and computing the requirements.
CO3	Plan, monitor, and manage project schedule, resources, and work assignments to ensure timely completion.
CO4	Perform professionally—as a team member, accepting responsibility, taking initiative, and providing leadership necessary to ensure project success.
CO5	Use formal and informal communications with team members and guide, make presentations and prepare technical document.
CO6	Provide sustainable solution within the context of legal framework addressing the societal and environmental concerns and upholding ethical issues.

CO-PO Mapping:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO1	PSO2	PSO3
CO1	3	3										3			
CO2		2	2		2							2	2		
CO3											2				
CO4									1						
CO5										1					
CO6						1	1	1							2



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Assessment Plan for CIE:

Tool	Remarks	Marks
Evaluation by Guide	--	48
Evaluation by Panel	--	50
Peer Evaluation	--	2
Total		50

Rubrics for Project Evaluation

Criteria	Exemplary	Proficient	Partially Proficient	Points
Problem Formulation (CO1)	Proposed an engineering problem that is relevant to current scenario, innovative, sustainable and practically achievable. Justification for the choice of problem selected. Has identified the functional and non-functional requirement.[15]	Proposed an engineering problem that is relevant to current scenario and practically achievable. Justification for the choice of problem selected. Has identified the functional and non-functional requirement. [6-14]	Proposed a fair engineering problem and practically achievable. Justification for the choice of problem selected. Has identified the functional and non-functional requirement. [1-5]	___/15
Planning (CO3)	Project schedule: Progress, milestones and deliverables with realistic estimates of the time (Gant chart). Identify resources: (hardware and software) required to accomplish the development effort. Cost estimates of hardware and software. [15]	Project schedule: Progress, milestones and deliverables (Gant chart). Identify resources: (hardware and software) required to accomplish the development effort. Cost estimates of hardware and software. [6-14]	Project schedule: Progress, milestones and deliverables are not planned accordingly. Identify resources: (hardware and software) required to accomplish the development effort. Cost estimates of	___/15



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

			hardware and software.[1-5]	
Literature Survey (CO1)	Exhaustive/substantial Literature Survey done. Reviewed credible/quality literatures. Clearly summarized the ideas of sources by a thorough review of academic literature. Publication of Literature Review paper. [25]	Reviewed credible/quality literatures. Clearly summarized the ideas of sources by a thorough review of academic literature. Publication of Literature Review paper.[15-24]	Reviewed few literatures. Fair summarization of the ideas of sources. Publication of Literature Review paper not done.[1-14]	___/25
High Level Design (CO2)	Design is as per problem formulation. Architectural design/ System design: Identifying the sub-systems. Abstract specification of the sub-systems, Interface design. Carefully chosen a methodology or approach that is well-suited to the formulated problem. [15]	Design is as per problem formulation. Architectural design/ System design: Identifying the sub-systems. Abstract specification of the sub-systems, Interface design. [6-14]	Design does not target the problem formulation optimally. Architectural design/ System design: Identifying the sub-systems. Abstract specification of the sub-systems, Interface design.[1-5]	___/15
Report (CO5)	Clear and Effective writing and adherence to appropriate report format. All figures, graphs, charts, and drawings are accurate, consistent with the text, and of good quality. They enhance understanding of the text. All are labelled correctly.[10]	Writing that is clear and effective for most part and minor errors in adherence to appropriate report format. All figures, graphs, charts, and drawings are accurate, consistent with the text, and of good quality. They enhance understanding of the text. All are labelled correctly.[6-9]	Unclear and ineffective writing and multiple errors in adherence to appropriate report format. All figures, graphs, charts, and drawings are accurate, consistent with the text, and of good quality. They	___/10



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

			enhance understanding of the text. All are labelled correctly.[1-5]	
Oral communication (CO5)	Clear and effective communication. Presentation includes appropriate contents and is clearly organized. Presentation highlights key ideas and closes with a strong conclusion. Answer questions/queries professionally. [10]	Communication is clear. Presentation includes appropriate contents and is clearly organized. Presentation highlights key ideas. Answer questions/queries professionally.[6-9]	Unclear communication. Presentation includes appropriate contents. Presentation highlights key ideas. [1-5]	___/10
Group Participation (CO4)	Did a full share of the work or more and volunteers to help others. Provided many good ideas; inspired others; clearly communicated ideas and needs. Completed assigned work ahead of time.[4]	Did almost as much work as others. Participated in discussions; on some occasions, made suggestions. Completed assigned work on time.[2-3]	Did less work than others. Listened mainly; Rarely spoke up, and ideas were off the mark. Needed much reminding; submission was late. [1]	___/4
Ethics (CO6)	Upholds the standards of honesty and integrity. Addressed the societal and environmental issues and responsibilities.[4]	Upholds the standards of honesty and integrity. Addressed few societal and environmental issues [3]	Upholds the standards of honesty and integrity.[1-2]	___/4
Peer Review (CO4)	To be evaluated by other team members			___/2
Total				___/100



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Semester	VI		
Course Title:	Web 3 Development		
Course Code:	23IC6AEWEB	Total Contact Hours:	30 hours
L-T-P:	0-0-1	Total Credits:	1

Introduction:

- This course introduces students to the principles and technologies of Web 3.0, focusing on decentralized applications (dApps), blockchain, smart contracts, and the integration of cryptographic methods.
- Students will develop hands-on experience by working on projects that utilize Web 3.0 technologies to build decentralized applications. Students can form groups of two to four for project work.
- The instructor will guide students in choosing relevant topics and throughout the project development process.

Course Outcomes

At the end of the course the student will be able to

CO1	Apply web 3 development technologies to solve real-world problems.
CO2	Develop decentralized applications using blockchain technology.
CO3	Design and implement smart contracts for various use cases.
CO4	Demonstrate teamwork and prepare documentation.

CO-PO-PSO mapping

	P O1	P O2	P O3	P O4	P O5	P O6	P O7	P O8	P O9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	3				3								3		
CO2			3		3	3	2	2				2	2		3
CO3			3		3			2					2		3
CO4									2	2					

Assessment Plan (for 50 marks of CIE)

Tool	Remarks	Marks
Internals	---	---
QUIZ	---	---
Lab Component	CIE through reviews	50
Alternate Assessment Tool	----	--
Total		50



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Rubrics for Project Evaluation:

Criteria	Excellent	Good	Satisfactory	Needs Improvement
Problem Identification & Relevance (10)	(10-8 Marks) Clearly identifies significant decentralized application opportunities with innovative solutions with concern to ethical issues.	(7-6 Marks) Recognizes relevant decentralized issues with practical solutions.	(5-3 Marks) Identifies basic issues with standard solutions.	(0-2 Marks) Fails to identify a relevant issue or solution.
Technical Implementation (10)	(10-8Marks) Exemplary implementation of Web 3.0 technologies showcasing efficiency, scalability, and technical excellence for sustainable real-world applications.	(7-6 Marks) Reliable and proficient technical performance meeting key objectives.	(5 Marks) Basic implementation incorporating essential features and functionalities.	(0-4 Marks) Inadequate or incomplete technical implementation.
User Experience & Interface (10)	(10-8Marks) Exceptional UI/UX design with emphasis on user-centric decentralized features and seamless interaction.	(7-6 Marks) Competent UI design focused on usability and functionality.	(5 Marks) Basic UI design with essential decentralized features.	(0-4 Marks) Poor or non-functional user interface lacking in user-centricity.
Group Participation (5)	(5 marks) Active engagement with exceptional collaboration and effective teamwork throughout the project lifecycle.	(4 marks) Consistent participation and constructive collaboration within the group.	(2-1 marks) Minimal but noticeable participation and occasional contributions.	(0 marks) Lack of active participation and collaboration in the group.
Presentation (5)	(5 marks) Professional, engaging presentation with outstanding visuals and	(4-3 marks) Well-structured presentation with clear content and effective	(2 marks) Basic presentation with some structure and varying delivery	(1 mark) Disorganized presentation lacking in coherence and adequate content.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

	comprehensive content demonstrating exceptional delivery skills.	delivery.	quality.	
Report & Documentation (10)	(10 marks) Comprehensive report covering all project aspects with meticulous documentation including methodology, design, and future scope.	(7 marks) Well-structured report with detailed coverage of project implementation.	(5 marks) Basic report with limited content covering essential project details.	(2-4 marks) Poorly structured and incomplete report lacking essential details.

Laboratory Plan:

Week	Activity	Content Deliverables by Assigned Teacher	Technologies/Skills to be Covered
1st	Formation of groups.	Introduction to Web 3 Technologies & Issue Identification	Overview of Web 3.0 and Blockchain technologies. Identifying issues for decentralized applications.
2nd	Project topic selection by each group. Presentation: Student and project topic introduction by each group.	Conceptualizing a Decentralized Application	Identifying problems and understanding decentralized needs. Tools for wireframing and prototyping (Figma, Sketch).
3rd	Design Layout of the Web Pages	Basic Front-end and Back-end Development for Decentralized Apps	Defining layouts based on project scope and objectives. Learning the basics of blockchain integration and decentralized web development. Front-end: HTML5, CSS3, JavaScript basics. Back-end: Introduction to Solidity, Ethereum, IPFS.
4th, 5th, and 6th	Front end and back-end implementation	Smart Contracts Development and Blockchain Integration	Techniques for creating smart contracts and integrating them into applications. Understanding blockchain networks like Ethereum, Polkadot.
7th, 8th, and 9th	Design and Development of dApps	Advanced Decentralized Technologies	Advanced front-end technologies for dApps, server management, and peer-to-peer networking.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

10th	Presentation by each group	Integrating Feedback & Refining Applications	Applying feedback from the mid-term review and refining the application for better performance and impact. Implementing feedback, optimization for performance, and security best practices.
11th	Complete Project Work Demonstration by each group	Final Project Presentations and Submissions	Students present their completed projects and submit their final work for assessment.
12th	Project Report Preparation		

Prescribed Text Book:

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1	Learn Ethereum	Xun (Brian) Wu, Zhihong Zou,	2 nd Edition	Dongying Song, PACKT PUB,	2023
2.	Mastering Blockchain	Imran Bashir	3 rd Edition	Packt Publishing ,ISBN: 9781839213199	2020

Tutorial Link:

Sl. No.	Title	URL
1.	The Road to Web3	https://www.web3.university/tracks/road-to-web3
2.	How To Create Your First Smart Contract	https://www.web3.university/tracks/create-a-smart-contract
3.	Learn Blockchain, Solidity, and Full Stack Web3 Development with JavaScript	https://www.youtube.com/watch?v=gyMwXuJrbJQ&t=17441s
4.	How To Build a Blockchain App with Ethereum, Web3.js & Solidity Smart Contracts	https://www.dappuniversity.com/articles/how-to-build-a-blockchain-app
5.	Getting Started with Web3 Development	https://www.udemy.com/course/getting-started-with-ethereum-solidity-development/?couponCode=SKILLS4SALEB



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

SEE Exam (50 Marks)

Evaluation of Projects carried out by students from the External examiner along with internal faculty.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Semester	VI		
Course Title:	Data Visualization using Tools		
Course Code:	23IC6AEDVZ	Total Contact Hours: 30 hours	
L-T-P:	0-0-1	Total Credits:	1

About the course: The course is designed to enhance programming and computation skills of students by exploring various features and extensive libraries of python programming language that are necessary for data science applications.

The students should work with a given dataset and create effective visualizations. The course will be executed in two cycles.

During Cycle 1, the students would be able to implement the key visualization techniques using Python tools like Matplotlib, Seaborn etc.

In Cycle 2, students will be exposed to industry-standard software tools like Tableau, Google Data Studio etc. to create compelling and interactive visualization of various types of data.

CIE 10 Marks, Lab Test 30 Marks and Report 10 Marks

Prescribed Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Python Data Science Handbook	Jake Vander Plas	Second Edition	O'Reilly	2017
2.	Pro Tableau: A Step by Step Guide	Seema Acharya , Subhashini Chellappan	Second Edition	Apress	2016

Reference Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Data Analysis and Visualization Using Python: Analyze Data to Create Visualizations for BI Systems,	Sossama Embarak		Apress	2018
2.	Python Data Visualization Cookbook	Igor Milovanović , Dimitry Foures , Giuseppe Vettigl	Second Edition	O'Reilly	2015



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

EBook

Sl. No.	Book Title	Author s	Edit ion	Publishe r	Ye ar	URL
1.	Data Visualization with Python and JavaScript	Kyran Dale	-	O'Reilly	2016	https://github.com/jllovet/dataviz-with-py-and-js
2.	Jumpstart Tableau: A Step-by-Step Guide to Better Data Visualization	Arshad Khan	-	Apress	2016	https://link.springer.com/book/10.1007/978-1-4842-1934-8

MOOC Course

Sl. No .	Course name	Course Offered By	Year	URL
1.	IBM Data Science	Coursera	2023	https://www.coursera.org/professional-certificates/ibm-data-science
2.	Data Visualization with Tableau	Coursera	2023	https://www.coursera.org/specializations/data-visualization

Laboratory Plan

Lab-cycle-1

1.	Using the sales_data.csv, create the visualization report for the following using Matplotlib:
a.	Get total profit of all months and show line plot with the following Style Properties: Generated line plot must include following Style properties: – 1. Line Style dotted and Line-color should be green 2. Show annotation 3. Add a square marker. 4. Add ticks for both X and Y axis
b.	Read Bathing soap facewash of all months and display it using the Subplot
2.	Using the sales_data.csv, create the visualization report for the following using Matplotlib:
a.	Get total profit of all months and show line plot with the following Style properties Generated line plot must include following Style properties: – 1. Line Style dashed and Line-color should be green 2. Show legend at the lower right location 3. Add ticks for both X and Y axis 4. Line width should be 2
b.	Read toothpaste sales data of each month and show it using a bar plot
3.	Using the sales_data.csv, create the visualization report for the following using Matplotlib:



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

a.	Calculate total sale data for last year for each product and show it using a Pie chart 1. Print the total sale inside each part 2. Explode the highest sale 3. Set the start angle=60
b.	Read face cream and facewash product sales data and show it using the horizontal bar chart
4.	Write a Python programming for the following:
a.	To display a horizontal bar chart of the sale of book. Use different color for each bar. 1. Sample data: Programming languages: Fict, Tech, Moti, Business, Nutri, Dev 2. Sale: 5.2,19.6, 8.7, 8, 7.7, 3.7 3. Add ticks for both axis 4. Show legend at the upper right corner
b	Write a Python program to create a stacked bar plot. Note: Use bottom to stack the women bars on top of the men bars. Sample Data: Means (men) = (22, 30, 35, 35, 26) Means (women) = (25, 32, 30, 35, 29) 1. Add labels and ticks 2. Use annotation
5.	Write a Python programming for the following:
a.	To create a pie chart with a title of the pass percentage of subjects. Sample data: Subjects: DSC, OOP, OPS, COA, MAT, Java Pass percentage (%): 40, 25.6, 8.8, 30, 7.7, 60.7 1. Print percentage inside the chart 2. Use explode property
b.	Using the sales_data.csv, read the total profit of each month and show it using the histogram to see the most common profit ranges
6.	Using the dataset planets.csv, create the visualization report for the following using Seaborn:
a.	Get the distance covered year-wise and show scatter plot with the following properties 1. Add “mass” as additional features 2. Use different markers 3. Control the range of marker areas with sizes
b.	Read the orbital_period of each year and show it using the histogram.
7.	Using the dataset planets.csv, create the visualization report for the following using Seaborn:



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

a.	Get the distance covered year-wise and show scatter plot with the following properties 1. Add “mass” and “method” as additional features 2. Change the default color palette 3. Display the complete legend
b.	Read the distance for each method and show it using the bar chart
8.	Using the dataset titanic.csv, create the visualization report for the following using Seaborn: 1. Demonstrate the use of “displot” 2. Plot the distribution using Kernel density estimation. 3. Use lineplot for any two suitable features 4. Generate scatter plot with different color palette
9.	Using the dataset titanic.csv, create the visualization report for the following using Seaborn: 1. Demonstrate the subplots (2x1) on scatter plots 2. Demonstrate the use of violin plot 3. Get different line plots for survival of passengers class wise. 4. Create visualization for strip plot without jitter
10.	Using the dataset titanic.csv, create the visualization report for the following using Seaborn: 1. Create a visualization using categorical plot and re-order the axis contents 2. Demonstrate the use of violin plot 3. Demonstrate the subplots (1x3) on line plots 4. Generate scatter plot with different color palette

Lab-cycle-2

I.	Create the visualization using Tableau for the “Corriander_seed_2021.csv” dataset. a. Demonstrate the use of filters (General, wildcard, condition and limits) b. Demonstrate the group creation, removing and renaming a group. c. Demonstrate the creation of constant set d. Create the visualization by using quick table calculation e. Customize the data using any three number functions
-----------	--



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

II.	Create the visualization using Tableau for the “Corriander_seed_2021.csv” dataset. a. Demonstrate the use of cascading filter, calculation filter and data source filter. b. Demonstrate creating Hierarchies c. Demonstrate the creation of computed sets d. Create a visualization using a calculated field e. Customize the data using any three string functions
III.	Create the visualization using Tableau for the “Corriander_seed_2021.csv” dataset. a. Demonstrate the use of cascading filter, calculation filter and data source filter. b. Demonstrate the group creation, removing and renaming a group. c. Create a visualization using a calculated field d. Customize the data using any three number functions e. Demonstrate the creation of constant set
IV.	Create the visualization using Tableau for the “supermarket_sales.csv” dataset. a. Demonstrate the use of filters (General, wildcard, condition and limits) b. Demonstrate the group creation, removing and renaming a group. c. Demonstrate the creation of constant set d. Create a visualization using a calculated field e. Customize the data using any three string functions
V.	Create the visualization using Tableau for the “supermarket_sales.csv” dataset. a. Demonstrate the use of cascading filter, calculation filter and data source filter. b. Demonstrate creating Hierarchies c. Demonstrate the creation of computed sets d. Create the visualization by using quick table calculation e. Customize the data using any three number function
VI.	Create the visualization using Tableau for the “supermarket_sales.csv” dataset. a. Demonstrate the use of filters (General, wildcard, condition and limits)



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

	b. Demonstrate creating Hierarchies c. Create the visualization by using quick table calculation d. Demonstrate the creation of constant set e. Customize the data using any three string functions
VII.	Create the visualization using Tableau for the “supermarket_sales.csv” dataset. a. Demonstrate the use of cascading filter, calculation filter and data source filter. b. Demonstrate the group creation, removing and renaming a group. c. Demonstrate the creation of constant set d. Create a visualization using a calculated field e. Customize the data using any three number functions

Course Outcomes

At the end of the course the student will be able to

CO1	Apply data transformations such as Joins, filtering, sorting, aggregation etc., for visualization using industry-standard software tools.
CO2	Design and create effective data visualizations using Python.
CO3	Identify opportunities for application of data visualization in various domains and communicate the results for documentation and interpretation.

CO-PO mapping

	PO 1	PO 2	PO3	P O4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO 1	3												3		
CO 2			3										3		
CO 3		3					3			3					

Proposed Assessment Plan (for 50 marks of CIE)

Assessment Tool	No. of Assessments	Marks
Internals	-	-
QUIZ/AAT	-	-
Lab Component	CIE + 1 Lab Test	50
Total		50



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Sem	VI		
Course Title:	Deep Learning		
Course Code:	23IC6PEDLN	Total Contact Hours:	40 hours
L-T-P:	3-0-0	Total Credits:	3

Unit No.	Topics	Hours
1	Introduction to Artificial Neural Networks: From Biological to Artificial Neurons: Biological Neurons, Logical Computations with Neurons, The Perceptron, The Multilayer Perceptron and Backpropagation, Regression and Classification MLPs, Implementing MLPs with Keras, Fine-tuning Neural Network Parameters. Introduction to Deep Learning: Deep Feedforward Networks, Gradient-based Learning, and Efficient Computation.	8
2	Convolution Neural Networks (CNN): The Architecture of the Visual Cortex, Convolutional Layer, Pooling Layer, CNN Architectures- LeNet-5, AlexNet, GoogLeNet, VGGNet, ResNet, Xception, Pre-trained Models for Transfer Learning, Classification and Localization.	8
3	Training Deep Neural Networks: Vanishing/Exploding Gradients, Reusing Pretrained Layers, Faster Optimizers, Avoiding Overfitting Through Regularization. Recurrent Neural Networks (RNN): Recurrent Neurons and Layers, Training RNNs, forecasting a Time Series, Handling Long Sequences-LSTM.	8
4	Representation Learning using Autoencoders: Stacked Autoencoders, Convolutional Autoencoders, Recurrent Autoencoders, Denoising Autoencoders, Sparse Autoencoders, Applications of Autoencoders, Transfer Learning, and Domain Adaptation. Generative Adversarial Networks (GANs): Difficulties of training GANs, Deep Convolutional GANs, Progressive Growing of GANs, StyleGANs.	8
5	Deep Generative Models: Boltzmann Machines, Restricted Boltzmann Machines, Deep Belief Networks, Deep Boltzmann Machines, Boltzmann Machines for Real-Valued Data. Applications: Computer Vision, Speech Recognition, Natural Language Processing.	8



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19
Autonomous Institute, Affiliated to VTU
DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Prescribed Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Deep Learning	Ian Goodfellow, Yoshua Bengio, Aaron Courville	1st	MIT Press	2016
2.	Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow	Aurelien Geron	2nd	O'Reilly	2020

Reference Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
1.	Deep Learning with Tensor Flow and Keras 3rd Edition	Amita Kapoor, Antonio Gulli, Sujit Pal	3rd	Packt	2022
2.	Learning Deep Learning: Theory and Practice of Neural Networks, Computer Vision, NLP, and Transformers using TensorFlow	Magnus Ekman	1st	Addison-Wesley Professional	2021

E-Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year	URL
1.	Deep Learning	Ian Goodfellow	First	MIT Press	2016	https://www.deeplearningbook.org/
2.	The Little Book of Deep Learning	François Fleuret	-	University of Geneva, Switzerland	2024	https://fleuret.org/public/lbdl.pdf

MOOC Course

Sl. No.	Course name	Course Offered By	Year	URL
1.	Deep Learning for Object Detection	Coursera	2024	https://www.mooc-list.com/course/deep-learning-object-detection-coursera



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

2.	Deep Learning	NPTEL	2024	https://onlinecourses.nptel.ac.in/noc24_ee04/preview
----	---------------	-------	------	---

Course Outcomes

At the end of the course, the student will be able to

CO1	Apply the fundamentals of deep learning algorithms for various complex problems
CO2	Analyze appropriate deep learning concepts based on the characteristics of the problem and the data.
CO3	Design and develop a model using deep learning to solve a problem.

CO-PO-PSO mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3												1		
CO2		2													
CO3			3		3				1	1					3

Proposed Assessment Plan (for 50 marks of CIE)

Assessment Tool	No. of Assessments	Marks
Internals	2	40
AAT	1	10
Total		50

AAT Plan:

Students should form a group of 3 or 4 to implement and analyze suitable deep learning techniques for a given real-world problem.

Rubrics:

Criteria	Exemplary	Proficient	Partially Proficient	Points
Implementation of the deep learning techniques using appropriate tools	(5-4) Implementation of the deep learning techniques has been done accurately.	(3-2) Implementation of the deep learning techniques has been done appropriately.	(1) Implementation of the deep learning techniques has been done.	____/4



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)

Testing for various cases	(1) The implemented model works for any given valid input.	(0.75) The implemented model works for almost all valid inputs.	(0.5) The implemented model works for any valid inputs.	___/1
Application/Relevance	(1) The designed technique has several applications and is relevant in the domain.	(0.75) The designed technique has few applications and is relevant in the domain.	(0.5) The designed technique has few applications and is not very relevant in the domain.	___/1
Report	(1) Clear and Effective writing and adherence to appropriate style guidelines	(0.75) Writing that is clear and effective for the most part and minor errors in adherence to appropriate style guidelines	(0.5) Unclear and ineffective writing and multiple errors in adherence to appropriate style guidelines	___/1
Oral communication (presentation)	(1) Clear and effective communication	(0.75) Communication is clear	(0.5) Unclear communication	___/1
Participation in Discussions	(1) Provided many good ideas; inspired others; clearly communicated ideas, needs, and feelings.	(0.75) Participated in discussions; on some occasions, made suggestions.	(0.5) Listened mainly; Rarely spoke up, and ideas were off the mark.	___/1
Total				___/ 10

SEE Exam Question paper format

In all five units, two questions are to be asked for 20 marks each as internal choice.



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF CSE (IoT & Cybersecurity including Blockchain)